THE JOURNAL OF COMPARATIVE NEUROLOGY 296:1-22 (1990)

Unbiased Stereological Estimation of the Number of Neurons in the Human Hippocampus

MARK J. WEST AND H.J.G. GUNDERSEN
Stereological Research Laboratory, University Institute of Pathology and Second University Clinic of Internal Medicine, Institute of Experimental Clinical Research, University of Aarhus, Denmark

ABSTRACT

The total numbers of neurons in five subdivisions of human hippocampi were estimated using unbiased stereological principles and systematic sampling techniques. The method addresses the problems associated with the results and conclusions of previous quantitative studies, virtually all of which have been based on biased estimates of neuron densities. For each subdivision, the total number of neurons was calculated as the product of the estimate of the volume of the neuron-containing layers and the estimate of the numerical density of neurons in the layers.

Each hippocampus was cut into 3-mm-thick slabs, transverse to the rostrocaudal axis. One 70-μm-thick section from each slab was used in the analysis. The volumes of the layers containing neurons in five major subdivisions of the hippocampus (granule cell layer, hilus, CA3-2, CA1, and subiculum) were estimated with point-counting techniques after delineation of the layers on each section. The numerical densities of neurons in each subdivision were estimated on the same sections with optical disectors. The sampling used in both estimates was performed systematically in all three dimensions.

In an example of five hippocampi, the mean numbers of neurons ($CV = SD/mean$) in the different subdivisions were as follows: granule cells 15×10^6 (0.28), hilus 2.0×10^6 (0.16), CA3-2 2.7×10^6 (0.22), CA1 16×10^6 (0.32), subiculum 4.5×10^6 (0.19). The stereological measurements contributed approximately 25% of the observed variance. Among the five subjects there was a significant inverse relationship between age (which ranged from 47 to 85 years) and the total number of neurons in CA1 (which ranged from 24 to 11×10^6).

An optimized sampling scheme for studies of the number of neurons in the human hippocampus has been designed on the basis of an analysis of variance of the estimates at different levels of the sampling scheme. Counting neurons in the five subdivisions of the human hippocampus with the optimized sampling scheme takes less than 4 hours.

Key words: aging, cortex, neurobiology, morphometry

The loss of neurons in the hippocampal region of the human brain has been the subject of a number of studies aimed at identifying morphological correlates of the cognitive changes associated with aging and and disease-related dementias. Although the results of these studies have often been discussed in terms of neuron loss, the data in virtually all of them are in terms of neuron density (i.e., neurons per unit volume, per section or per standardized segment), in some cases corrected for section shrinkage or brain size; (for reviews see Mani et al., '86; Coleman and Flood, '87; Flood and Coleman, '88). There are two aspects of the data of the earlier studies that confound the conclusions that can be drawn from these studies and have led to considerable dis-

Accepted November 20, 1989.

Address reprint requests to Mark J. West, Stereological Research Laboratory, Bartholin Building, University of Aarhus, DK-8000 Aarhus C, Denmark.

© 1990 WILEY-LISS, INC.

agreement about the role of neuron loss in the memory deficits associated with aging and diseases such as Alzheimer's disease (Coleman and Flood, '87).

The first is that the estimates of numerical density in the earlier studies have been obtained from two-dimensional images or probes and are biased because the probability that a neuron or, as is more commonly estimated, its nucleus, appears in a section is related to its size, shape, and orientation. For example, larger nuclei will appear more often in sections than will smaller nuclei, and nuclei will appear more often in sections when they are cut across their long axis. Biases of this type are essentially irreversible and undetectable in the data themselves. Additional biases are introduced by implicit and explicit assumptions about size, shape, and orientation that accompany corrections for split nuclei, (Abercrombie, '46; Floderus, '44), lost caps, and overprojection (Holmes, '27).

The second and much more problematic aspect of the data of earlier studies is that only the number of neurons per unit volume of tissue is reported, and nothing is known about the volume of the region or, consequently, about the total number of neurons. The importance of knowing the volume of the structure (the reference volume) in which estimates of neuron density is performed is illustrated in the work of Swaab and Uylings ('87) and Pakkenberg and Gundersen ('88). In both of these studies, the numerical density within a well-defined region was unchanged, whereas the reference volume and, as a consequence, the total number of neurons were significantly different. The product of neuron density and reference volume, the total number of neurons, is a far superior parameter in terms of scientific effectiveness (for further discussions of densities vs. total number see Brændgaard and Gundersen, '86; Gundersen, '86). Only when the reference volume is *known* can neuron density be used in a meaningful discussion of the functional capacity of a neural structure.

In the belief that absolute quantities, such as the total number of neurons in a well-defined region, can be of value when studying the structural dynamics of the brain, we describe in the present paper how two recent developments in stereology can be used to estimate the number of neurons in the subdivisions of the human hippocampus. The first of these is a three-dimensional probe, the disector (Sterio, '84), which can be used to obtain unbiased estimates of neuron density. It is free of assumptions about size, shape, and orientation of the entities being counted and unaffected by lost caps, split nuclei, and overprojection due to section thickness. The second is the application of systematic sampling (as, for example, described by Gundersen and Jensen, '87) to estimates of neuron density made with the disector and estimates of the reference volume made with point-counting techniques. In the present paper, we describe a sampling scheme that has been optimized for estimating neuron density and reference volume (and therefore the total number of neurons) in the different subdivisions of the human hippocampus. The estimation procedure is intuitively simple, readily performed, and requires little special equipment beyond that found in well-equipped histology laboratories.

GENERAL MATERIAL AND METHODS

Total number of neurons

The borders of the neuron-containing layers of the subdivisions of the hippocampus are defined on a relatively small number of sections (~10) taken systematically along the entire length of the hippocampus. On these sections the simply performed, highly efficient Cavalieri estimator (Cavalieri, '66) is used to estimate the reference volume $V(ref)$ from the areas of the sectional profiles of the layers and the known distance between the sections. The three-dimensional disector probe, with its unbiased counting rule, is used systematically within the delineated regions to estimate the numerical density (N_V) of neurons. The product of the reference volume and the numerical density is an unbiased estimate of the total number of neurons.

Unbiased estimates of $V(ref)$ and N_V

The methodology employed here gains its strength from the unbiased nature of the estimators of reference volume $V(ref)$ and numerical density (N_V) and the systematically random sampling (i.e., with a fixed and known periodicity from a random starting point) that is carried out in all three dimensions at all levels of the sampling scheme (sections, disectors, and points). Because the estimators are unbiased, they approach the true value of the parameter as one increases the number of samples at the appropriate sampling level. Without *random* sampling there is no way to obtain *unbiased* estimates. The use of random systematic sampling, as opposed to random independent sampling, increases considerably the efficiency with which one approaches the true value of the parameter being estimated.

Rationale of the sampling scheme

The design of a rational and efficient sampling scheme for estimating the number of neurons in the hippocampus of a group of individuals was based on the following considerations. The overall sampling scheme may be viewed as a hierarchy of sampling levels, each of which contributes to the observed overall variation of the final estimate: the mean total number of neurons in a particular subdivision of the hippocampus in normal man. At the uppermost or first level is a group of individual human hippocampi. Because of ordinary biological variation, the number of neurons in the hippocampi of different individuals will vary. Most importantly, for a given group of individuals, the biological variation is *fixed*, with the consequence that the biological variation determines the upper limit for the precision needed in the estimates of total number in a particular hippocampal subdivision of an individual. Improvements in the precision of the estimate beyond this point require extra effort but will *not* provide more or better information.

At the next sampling level, each hippocampus is sectioned systematically into a number of sections. Because the areas of these sections are not identical, estimates of total volume will show a certain variation if sampling and estimation were repeated in the same hippocampus; i.e., the estimates have a certain limited precision. The same is true for the estimates of average numerical density because of inhomogeneities in the packing densities of neurons within the layers. There are a number of important points to be made here with regard to the precision of these estimates.

From the observations made on a set of sections, it is possible to *estimate* the precision of the estimates made in an individual. This precision is most conveniently expressed as the coefficient of error (CE) of the estimate. The strictly systematic nature of the sampling scheme with n consecutive sections means, however, that one cannot calculate the CE in the usual way: $CE = CV/\sqrt{n}$, where the coefficient of

variation, $CV = SD/mean$, is among the n sections. This formula is valid *only* if the n observations are *independent*, which most obviously is not the case for consecutive sections. Fortunately, Matheron ('71) has developed the necessary theory for obtaining robust estimates of the CE for systematic samples of this type. This theory has been expanded and illustrated in a stereological context by Gundersen and Jensen ('87). (A detailed example of how one estimates the CE of the Cavalieri estimate of total volume from sections taken systematically along one axis is presented below in Table 3.) As usual, the CE is an inverse function of the number of samples (sections) from the object. However, with systematic sampling, the CE declines much faster than by a factor of $1/\sqrt{n}$ as it does for *independent* observations. Typically, the CE of a systematic sample is proportional to $1/n$ (cf. Fig. 8 in Gundersen and Jensen, '87), an important point when designing and optimizing the sampling scheme.

The price to pay for this valuable statistical property is that it is a bit more complicated than usual to estimate the CE, and the estimate is only an approximation, although a good one (Gundersen and Jensen, '87). This seems a small price to pay, however, because the only alternative is to use a completely random scheme with truly independent sampling. Such a scheme would require more sampling to obtain the desired CE and also would require that the hippocampus be glued together and resectioned after each new section was selected.

Below the level of sections there are still two more sampling levels: fields of vision sampled on each section and points hitting or neurons counted in each field of vision. However, one of the advantages of the approach used here is that one need analyze the sampling scheme only to a depth just below that which contributes most to the overall variance. Because the largest contribution will almost always be at the uppermost level, i.e., among individuals, it is generally necessary to analyze only to the level of sections. (For a more detailed description of the analysis of stereological sampling schemes for the estimation of total quantities in neuroscience, see Gundersen and Jensen, '87; Pakkenberg and Gundersen, '88; Brændgaard et al., '89, and references therein.)

Because the variance of the estimates made at the different levels of the sampling scheme contribute to the variance of the estimate of neuron number N, the guiding rule for optimal sampling is to put most effort into the level that contributes most to the overall variance. For adjacent sampling levels, the observed relative variance OCV_j^2 at the higher level is in general the sum of the true relative variance CV_j^2 at that level and the contribution OCE_{j+1}^2 from the relative observed variance at the level below:

$$OCV_j^2 = CV_j^2 + OCE_{j+1}^2$$

OCV = observed relative variation, observed SD/mean
CV = true relative variation, true SD/mean
OCE = computable variation of the stereological estimate, i.e. the observed coefficient of error, SEM/mean

When the observable, computable variance, OCE_{j+1}^2, of the stereological estimate at the lower level is half or less than that of the observed variance at the higher level, OCV_j^2, sampling at the lower level should be terminated because the true variation at the higher level, CV_j^2, is then contributing most to the observable variation at the higher level.

At the highest levels of the sampling scheme, this means that the observed and computable variation OCE^2 of the estimate of N for an individual should be less than half that of the observed variance among individuals, $OCV^2(N)$, because the major contribution to the latter is then the true biological variance, CV^2. The variation at the more important highest level (i.e., among individuals, CV) is unknown when a structure is studied for the first time. It is therefore necessary to make educated guesses and work backwards after a sample at the highest level has been obtained in order to make an evaluation of the sampling efficiency. At the start of the present analysis, it was assumed that the CV would be of the same magnitude as that for the number of neurons in human neocortex, at least 10 to 15% (Brændgaard et al., '90). As a consequence, a CE of about 0.10 for the estimate of N in an individual was considered a rational starting point for this methodological study.

MATERIALS

The brains of five males, aged 47, 56, 66, 80, and 85 years, were procured in accordance with the laws of Denmark governing the postmortem use of human tissue. None of the patients had shown signs of neurological disorder prior to death. Their data and causes of death are shown in Table 1. The entire brains were fixed in 4% formalin for 6–12 weeks after autopsy, at which time the basal half of the temporal lobe, including the hippocampus, was removed from each brain.

DETAILED METHODS

Sectioning and histology

The portion of the temporal lobe that was removed was embedded in 6% agar, and 3-mm-thick parallel slabs were cut in the frontal plane (Fig. 1) with a macrotome (Gundersen and Jensen, '87, Fig. 24). The most rostral slab collected contained the most rostral extent of the genu of the hippocampus. The most caudal slab contained the gyrus fasciolaris. Slabs containing the subsplenial gyrus were not included in the analysis. The position of the first cut with respect to the rostral end of the hippocampus was random within the first 3-mm interval. Cutting the hippocampus in this manner resulted in 12–15 slabs, each 3 mm thick, from each hippocampus.

Each 3-mm-thick slab was embedded in glycolmethacrylate, (hydroxyethylmethacrylate, Technovit 7100, Kulzer and Co., Wehrheim, Federal Republic of Germany) with the rostral face up. Prior to embedding, the slabs were dehydrated through a graded series of ethylalcohol solutions (70%, 70%, 96%, 96%, 99%, 99%, for 75 minutes each) and infiltrated with the resin, without hardener, for 6 days, with daily changes of the resin. One 70-μm-thick section was cut from each embedded slab with an LKB Historange microtome and stained with a Giemsa stain (Iniquez et al., '85) modified for use with glycolmethacrylate sections. The sec-

TABLE 1. Information About Patients From Whom the Hippocampi Used in This Study Were Obtained

Patient no.	Age (years)	Brain weight (g)	Body weight (kg)	Height (m)	Cause of death
1	47	1,559	74	1.83	Malignant lymphoma
2	56	1,418	64	1.74	Peritonitis
3	66	1,350	73	1.72	Bladder cancer
4	80	1,338	54	1.71	Myocardial infarction
5	85	1,412	85	1.80	Acute bronchitis

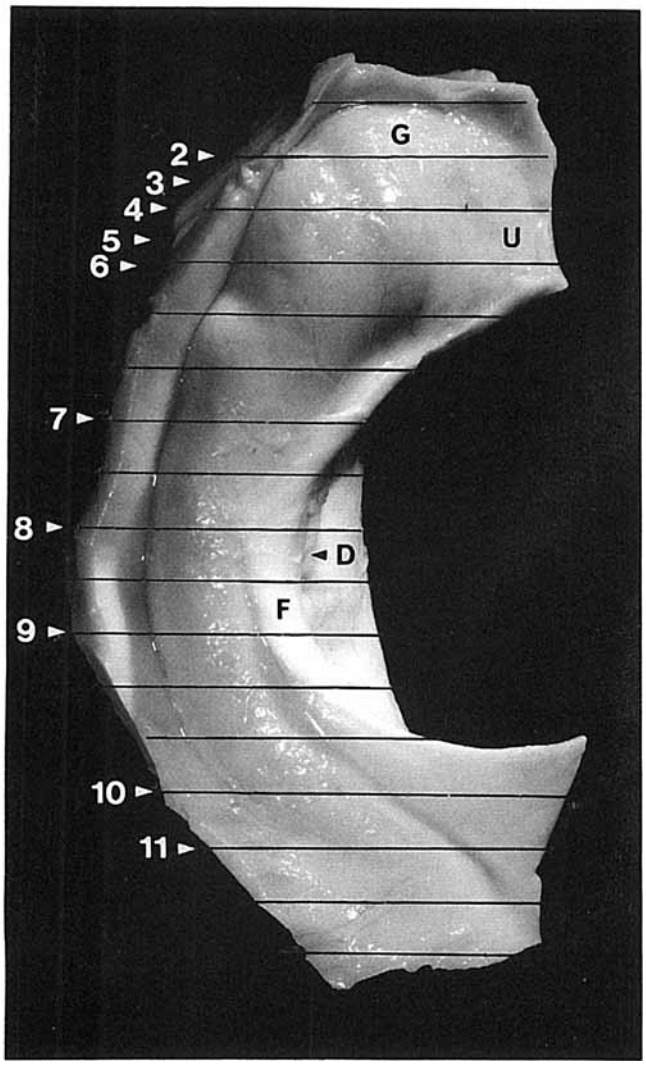


Fig. 1. Dorsal view of the left human hippocampus showing the position of cuts in the frontal plane (black solid lines) used to divide it into 3-mm-thick slabs along its entire rostrocaudal extent. The rostral pole is at the top, medial to the right. The numbers at the left indicate the numbers of the figures in which sections from specific levels are depicted. G, genu; U, uncus; D, dentate gyrus; F, fornix.

tions were floated on a cold-water bath after sectioning with 38-mm-wide glass knives (Ralph knives), mounted on ordinary glass slides, and taken immediately to an 60°C oven for drying to insure that the rather thick sections adhered to the slide during processing and staining. The staining solution was made of 10 ml Giemsa stain stock solution (Struers A/S, Copenhagen, product 310113-104), 40 ml of distilled water, and 2 drops of 1% acetic acid. The mounted sections were placed in the staining solution for 70 minutes at room temperature, rinsed in 1% acetic acid for 10 minutes, and differentiated in 96% ethanol for 10 minutes and 99% ethanol for 15 minutes. The sections were mounted with a cover class with Eukitt mounting medium.

Prior to initiation of the embedding procedure, the areas of nine arbitrarily chosen slabs from three hippocampi were determined by point counting under a dissection microscope with a quadratic grid with an interpoint distance of 1 mm. After embedding, sectioning, and staining, the areas of the sections from these slabs were again estimated to determine the extent of tissue shrinkage and section compression during the preparation of the sections.

Definitions of the neuron-containing layers of the subdivisions of the hippocampus

The layers containing neurons in each of the five major subdivisions of the hippocampus were defined as described below. The layers were 1) the granule cell layer of the dentate gyrus, 2) the hilus of the dentate gyrus (roughly identical with CA4, see below), 3) the pyramidal cell layer of CA3-2, 4) the pyramidal cell layer of CA1, and 5) the cellular layers of the subiculum. The boundaries of these layers are illustrated at various levels along the rostrocaudal axis of the hippocampus in Figures 2–11. The sections in these figures have been photographed at a constant magnification and have been selected with a higher frequency at rostral levels, where changes in the two-dimensional organization of the subdivisions are more dramatic. These sections (Figs. 2–11) are not equally spaced along the axis as are those selected for the analysis of neuron number (black lines, Fig. 1). Each section is accompanied by a diagram depicting the delineations of the layers of the subdivisions in which the total numbers of neurons were estimated.

Granule cell layer of the dentate gyrus

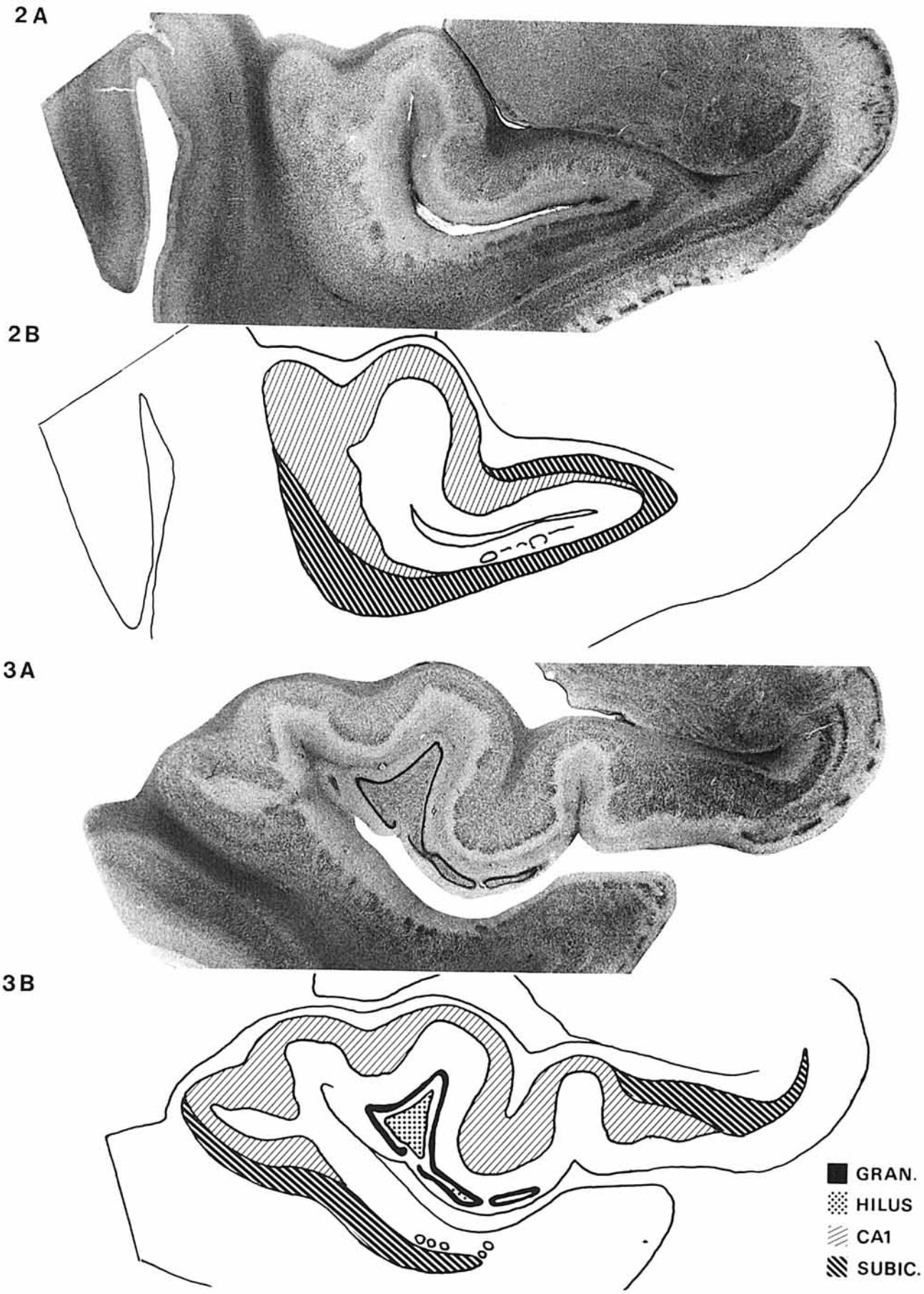
The granule cell layer is a thin, undulating layer of tightly packed neuron cell bodies. It is readily identified in all sections in which it is present because of the intensity with which it stains and the fact that it is not continuous with the cellular layers of the other subfields (Fig. 12). Because of the folding of the layer, both along the body of the hippocampus and in the digitations of the head and tail, islands of granule cells can be found in a multitude of forms (Figs. 8, 9), but they are identified unequivocally on the basis of the intense staining of this layer. Individual variations in the number and size of the digitations (Gertz et al., '72) present more difficulties in the delineation of the other cellular layers than they do in the delineation of the granule cell layer.

Hilus of the dentate gyrus

In the hilus there is evidence of cell lamination, best appreciated at levels where the section plane approximates the transverse plane of the hippocampus (Fig. 12A, B). The part of the hilus for which the volume was estimated and in which the neuron density was estimated included the outer hilar cell layer (OCL), the inner plexiform layer (IPL), and the inner hilar cell layer (ICL) (Geneser, '87), corresponding to Lorente de Nó's ('34) CA4. The boundary between the

Fig. 2. **A:** Frontal section through the rostral hippocampus at a level corresponding to 2 in figure 1. This section is the first of a series (Figs. 2–11) that illustrate variations in the organization of the neuron containing layers of the different subdivisions along the rostrocaudal axis of the hippocampus. Medial is to the right. At this level the hippocampus is composed of CA1 and subiculum. **B:** Diagram of the neuron containing layers of the subdivisions in the section shown in A. The key to the shading can be found in Figure 3B.

Fig. 3. **A:** Frontal section through the rostral hippocampus at a level corresponding to 3 in Figure 1. At this level, the hilus and granule cell layer begin to emerge. **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A.


2 A
2 B
3 A
3 B
GRAN.
HILUS
CA1
SUBIC.

pyramidal cell layer of CA3 and hilus can be readily identified as the point at which the tightly packed layer of CA3 pyramidal cells appears to bend back upon itself and to become more diffusely organized as the inner hilar cell layer (Fig. 12B). The inner limit of the hilus is formed by two concave borders: one along the deep edge of the inner cell layer, from the transition of the pyramidal cells in CA3 to CA4 toward the tip of the subpyramidal blade of the granule cell layer and the other from the CA3–CA4 pyramidal cell transition toward the suprapyramidal blade of the granule cell layer (Fig. 12B). The outer limit of the hilus was defined as *the outer border of* the outer cell layer. The outer plexiform layer (OPL) was excluded from the part of the hilus that was quantified because of the paucity of neurons in this layer. The inner plexiform layer was included in the definition because the two cellular layers (OCL and ICL) were not separable at all levels of the section series but were assumed to embrace an inner plexiform layer at all levels.

Pyramidal cell layers of the hippocampus proper

The pyramidal cells of the hippocampus were divided into two zones, one lying near the dentate gyrus, which contained large, tightly packed perikaria and corresponds to CA3 and CA2, and another lying closer to the subiculum, which contained smaller pyramidal cells dispersed along the radial axis in a thicker but less densely packed layer corresponding to the pyramidal cells of CA1. The border between the two zones was readily defined by the abrupt transition in cell organization and size (Fig. 13). The zone containing the pyramidal cells of CA1 expands progressively in the direction of the subiculum and is organized in two layers, the most superficial and densely packed of which overlaps the subiculum creating an oblique border with the latter (Fig. 14).

Subiculum

The subiculum contains a thick zone of pyramidal cells organized in two layers. The external layer contains large pyramidal cells, whereas the internal layer contains medium-sized pyramidal cells. The internal layer extends farther in the direction of CA1 and is overlapped by the superficial layer of CA1 pyramidal cells in the area subicularis lateralis marginalis (Braak, '80). The border between CA1 and subiculum is consequently oblique rather than perpendicular to the cortical surface. The border of the subiculum with the presubiculum is also oblique but tilted in the opposite direction. The small characteristic pyramidal cells of the presubiculum overlap the superficial edge of the external pyramidal cell layer of the subiculum in the area subicularis medialis (Braak, '80). The two oblique borders give the subiculum the appearance of a trapazoid with its base toward the white matter (Fig. 14). Isolated clouds of presubicular neurons appear regularly in the plexiform layer of the subiculum (Fig. 14, right arrow). These were not included within the borders of the subiculum. When cell counts were made, no attempt was made to distinguish between the pyramidal and the stellate cells that are found throughout the layers of the subiculum.

No attempt was made to identify a separate prosubiculum. In sections in which the smaller pyramidal cells characteristic of this zone (Rosene and Van Hoesen, '87) could be identified in the superficial layer of the transition between CA1 and subiculum, they were included in CA1 (Fig. 14, left arrow). In the rostral, dorsal, medial part of the uncus, a subfield with unique cellular characteristics could be identified, the hippocampal-amygdaloid transition area (HATA; Fig. 5). In spite of histochemical and hodological similarities with the subiculum (Rosene and Van Hoesen, '87), this subfield was treated as a unique subfield and not included in any of the zones that were quantified.

Delineation of the neuron-containing layers

The delineations of the neuron-containing layers were made under a dissection microscope at 10× on the underside of the slides with an ultrafine, felt-tipped, waterproof marking pen. This created little problem with parallax when viewed in a conventional microscope and prevented dissolution of the ink when immersion oil was used with the optical disectors. Because the line used to define the border between adjacent subdivisions was of positive thickness, a convention was established to use one of the edges of the line as the boundary. In this way, it was ensured that the sum of the individual areas of the neuron-containing layers was the same as the total area of the neuron-containing layers.

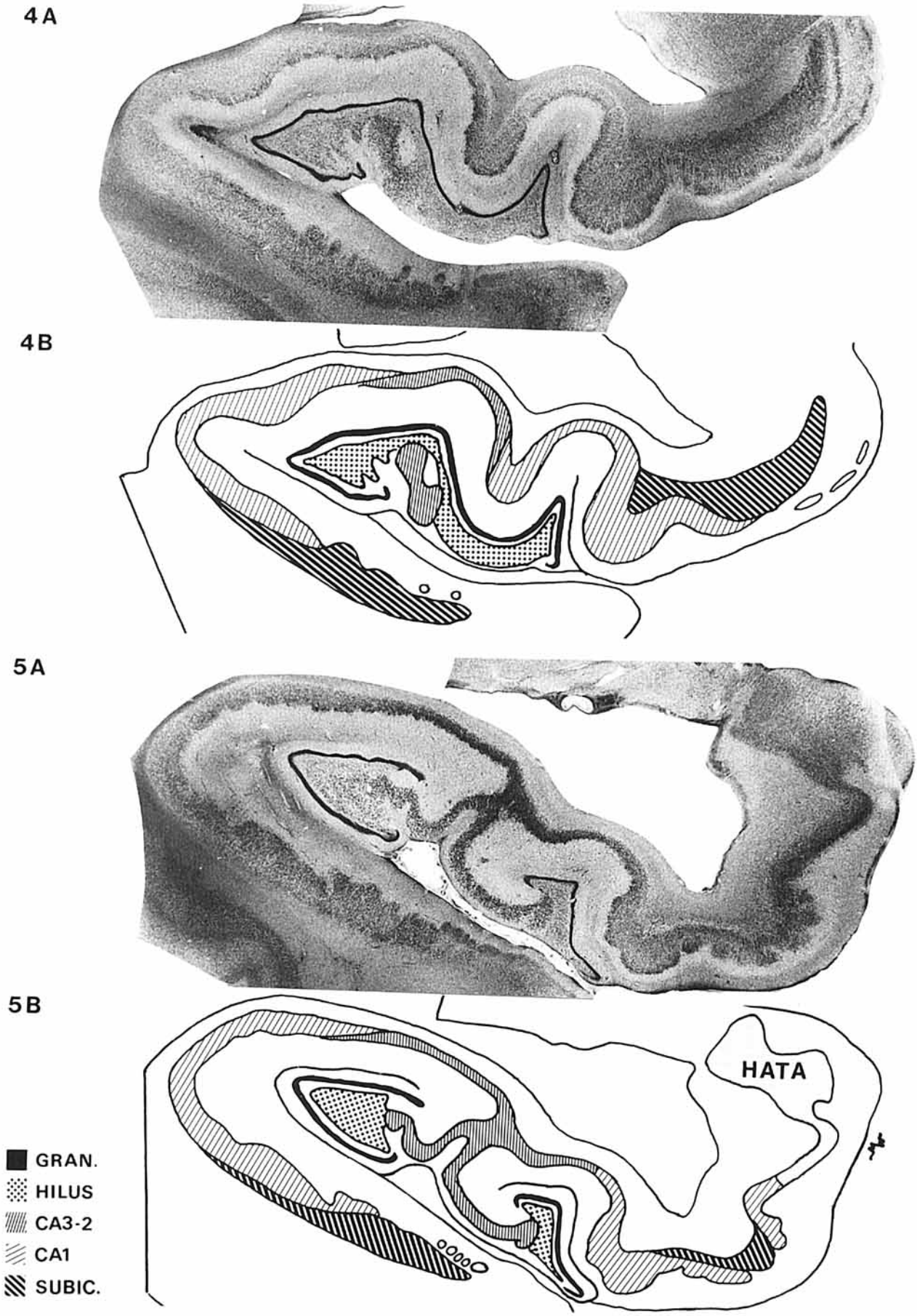
A number of practices evolved during the course of the study that facilitated the delineation process. Although 70-μm-thick sections are not required for the stereological analysis, it was found that sections of this thickness provided the architectonic information necessary for the delineations. It was also found that by starting the delineation process on the sections at the more central levels, where the hippocampus is cut in the transverse plane (Fig. 7), and working systematically in the rostral direction toward the more complicated genu and uncus, the architectonic and topographic information from the prior sections was helpful in making the delineations in the more rostral sections. The topographic relations of the subdivisions given in Figure 306 of Stephan's book on the allocortex (Stephan, '75), the detailed descriptions of the genu and uncus of the monkey by Rosene and Van Hoesen ('87) and of man by Duvernoy ('88), and the illustrations of the serially sectioned human subiculum by Braak ('72) were found to be valuable aids when delineating the layers in the rostral parts of the hippocampus.

Estimates of the reference volume *V(ref)* of the neuron-containing layers of hippocampal subdivisions

According to the classic mathematical principle of Cavalieri ('66), an unbiased estimate of the volume V of an object can be obtained from the sum of the areas (a_i) of the individual profiles of the object on a set of n systematically posi-

Fig. 4. **A:** Frontal section through the hippocampus at a level corresponding to level 4 in Figure 1 showing the multiple representations of the various subdivisions produced by the bending of the rostral hippocampus back upon itself. **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A.

Fig. 5. **A:** Frontal section through the hippocampus at level 5 of Figure 1 showing the apparent separation of uncal components (right) from those of the body of the hippocampus. **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A. HATA, hippocampal-amygdaloid-transition-area.


4A
4B
5A
5B
HATA
GRAN.
HILUS
CA3-2
CA1
SUBIC.

tioned parallel sections through the object that are separated by a known constant distance, t.

$$V = t \cdot \sum a_i$$

The only other condition required to ensure that the estimator is unbiased is the random placement of the first section plane within the first interval length t.

It was decided a priori that a coefficient of error (CE) for the estimates of the volumes of neuron-containing layers of the subdivisions of a hippocampus should be <0.10 to be compatible with the expected interhippocampal variation in neuron number. This level of confidence in the estimate can be achieved with about ten sections when the areas of the profiles of the layers are measured with an accuracy of about 10% (Gundersen and Jensen, '87).[1] From pilot studies, it was determined that 9–15 sections would be obtained through the different subdivisions of the hippocampus if frontal sections were separated by 3 mm.

With the exception of the granule cell layer of the dentate gyrus, the areas of the profiles of the layers on the sections used for the volume determinations were estimated by counting the points (P_i), in a quadratic lattice placed directly on the slide, that hit the profiles. The number of points hitting the $i'th$ profile is a direct and unbiased estimate of its area a_i

$$a_i = \frac{a}{p} \cdot P_i$$

where p/a is the number of points (p) per unit area (a) in the lattice. It has previously been shown, for objects with shapes similar to those of the cellular layers of the hippocampus (Gundersen and Jensen, '87), that not more than 50 to 100 counts (hits) are required on the entire set of sectional profiles to produce a CE of less than 0.10. Pilot studies were performed to determine what the interpoint distance of the point lattices should be so that 50 to 100 points would hit a given set of profiles. This resulted in the use of a lattice with an interpoint distance of 1 mm for hilus, CA3-2, and subiculum and 2 mm for CA1. The two lattices were combined on one transparent sheet, as shown in Figure 8B. The counting was done under a dissection microscope at a magnification of 10×.

Volume of the dentate granule cell layer

The boundaries of the layers used for estimating volume by point counting must have exactly the same definitions as those used for cell counting. Because the granule cell layer is relatively thin, a final magnification of at least 300× is required to define its boundary. At this magnification, it is not practical to count points in a grid placed physically on the section. Moreover, the areas of very long, thin profiles are not estimated efficiently with quadratic lattices. As a consequence, a different approach involving the superimposition of a "linear" point-counting grid on a video image of the section was used.

Because the entirety of the layer could not be viewed in the same field at the magnification required for defining the boundaries, it was necessary to analyze sequentially portions of the region of the section that contained the granule cell layer. This was accomplished by superimposing a linear array of six points, generated by a microcomputer, onto a video image of the section. (See rectangle to the lower right in Figure 12C, depicting the size of the video image and the six-point array relative to the size of the granule cell layer.) The section was moved with programmable stepping motors in the direction of the row of points so that the ends of the consecutively positioned arrays roughly coincided. When the region of the section that contained the granule cell layer moved out of view, the section was moved a different predetermined distance, perpendicular to the last row, and a new row of fields of view, parallel to the previous one, generated in the opposite direction through the region containing the layer. The first row was randomly positioned well outside the layer. Consecutive parallel rows were scanned until they no longer passed through the granule cell layer. This process is shown diagramatically in Figure 12C.

To maximize the efficiency of this procedure, the scanned rows were oriented roughly perpendicular to the mediolateral axis of the layer so that a maximal number of arrays passed through the layer. This was accomplished with an Olympus microscope object rotator (manufactured by BICO, Copenhagen), a slide holder that could be rotated independent of the x, y movement of the microscope stage (see Fig. 9 in Gundersen et al., '88).

The interpoint distance of the array was dimensioned so that, on the average, one point hit the layer whenever the row of points intersected the layer. The distance between rows was dimensioned so that the total number of hits (i.e., hits on all the ~ten sections) was ~100. This resulted in the use of an array with an interpoint distance of 0.092 mm, a vertical field-to-field displacement of 0.552 mm and an interrow displacement of 1.33 mm.

Because the displacement of the six-point array is known from the systematic movement made on the x- and y-axes, the area associated with each point, a/p is known:

$$\frac{a}{p} = \frac{\text{horizontal displacement} \times \text{vertical displacement}}{6}$$

$$= 1.33 \times 0.552/6 = 0.122 \text{ mm}^2$$

Although the rough borders of the granule cell layer were delineated with a felt-tip pen at a low magnification under a dissecting microscope, these borders were used only for orientation purposes during the displacement of the point arrays. The precise definition of the border of the layer was made when the layer was viewed at the magnification of 300× used for point counting. Only granule cells with cell bodies that appeared to be in immediate contact with the layer of contiguous cell bodies were included within the borders.

Estimates of N_V with the optical disector

Estimates of N_V in the neuron-containing layers of the different subdivisions were made from systematic disector samples. In its most general form, valid for both opaque and transparent sections, the disector (Sterio, '84) consists of two parallel planes (hence its name) separated by a known distance that is less than the height of the particles to be

[1]Although it is important that the areas of the sectional profiles of the layers be estimated with a certain precision, the most important variable affecting the precision of the volume estimate is the *number* of sections. No explicit attempt was made to estimate the precision of the individual area measurements, although it can be assumed that the precision was of the magnitude of 10% (see the nomogram for CE of area estimates using systematic points in Figure 18 in Gundersen and Jensen, 1987).

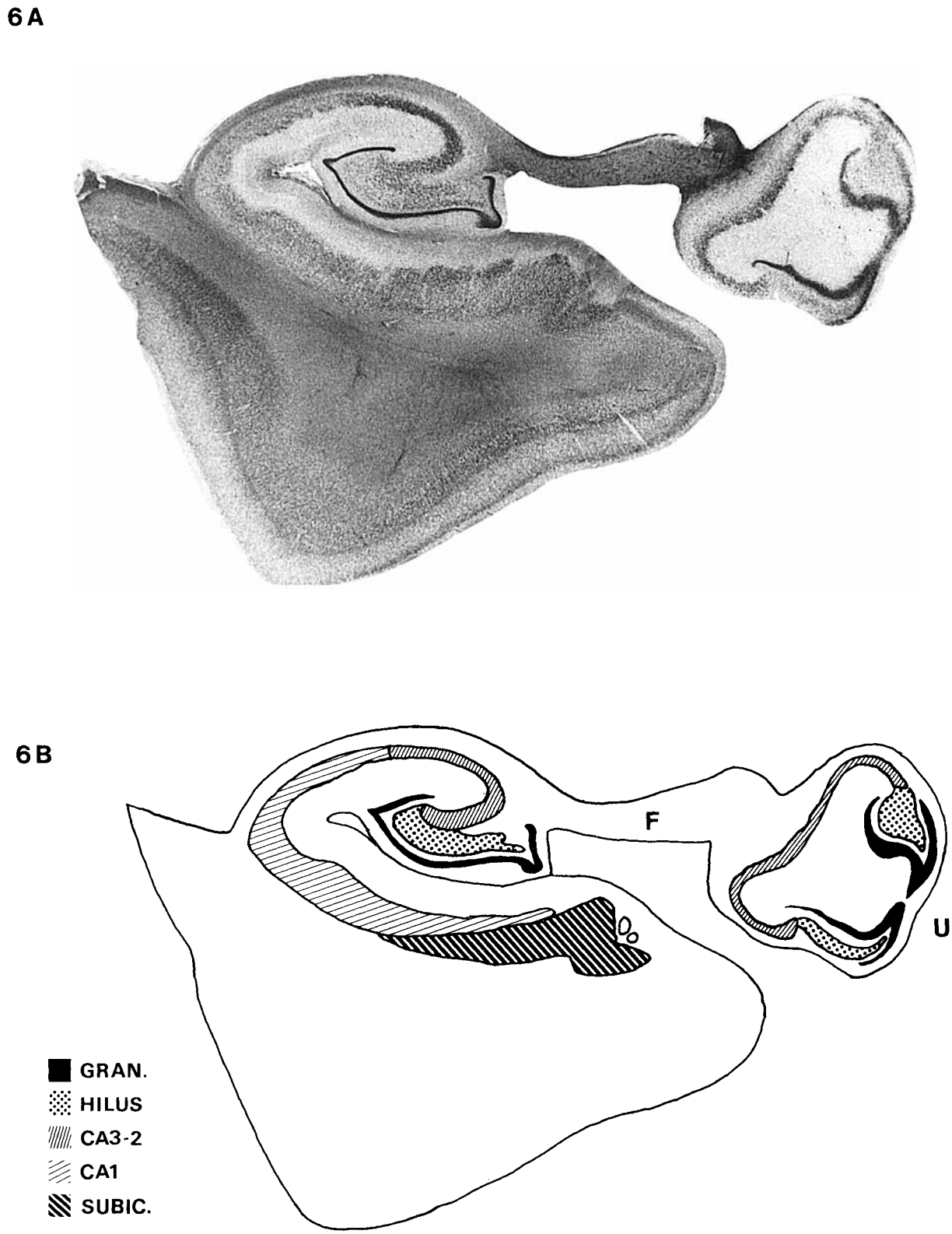
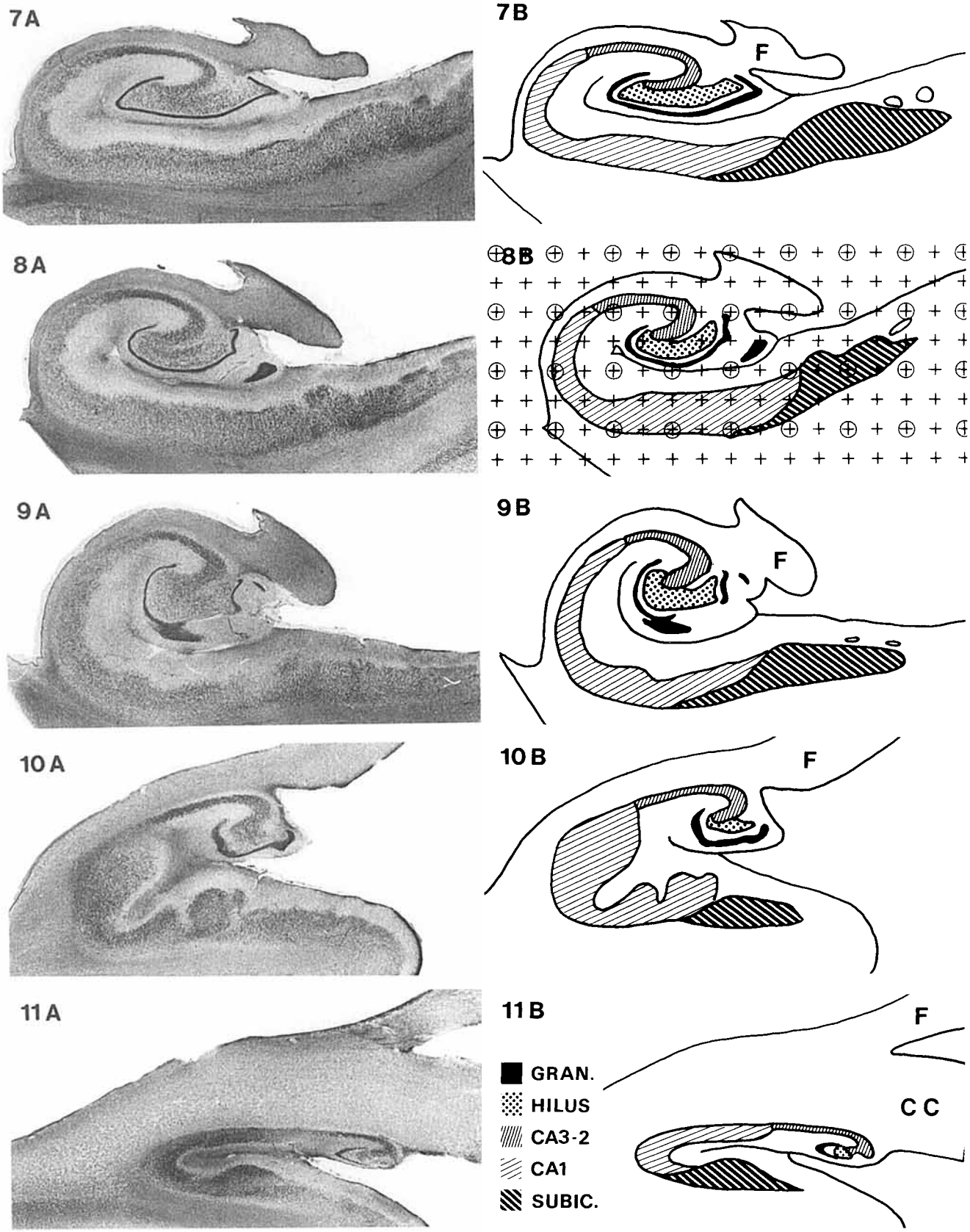


Fig. 6. **A:** Frontal section through the hippocampus at level 6 of Figure 1. **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A. F, fornix; U, uncus.

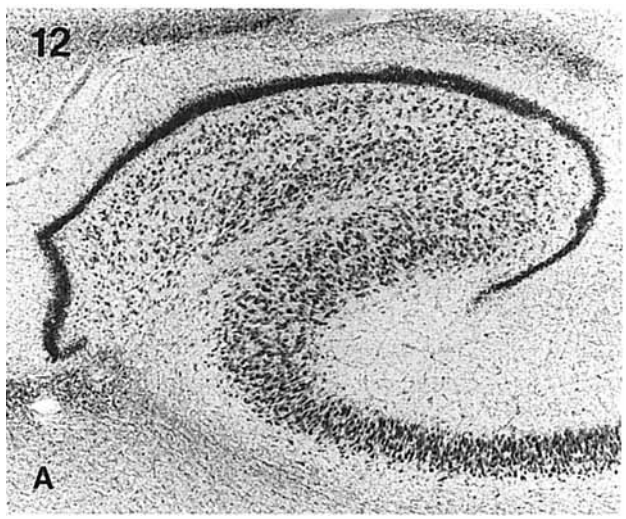

7A
7B
F
8A
8B
9A
9B
F
10A
10B
F
11A
11B
GRAN.
HILUS
CA3-2
CA1
SUBIC.
F
C C

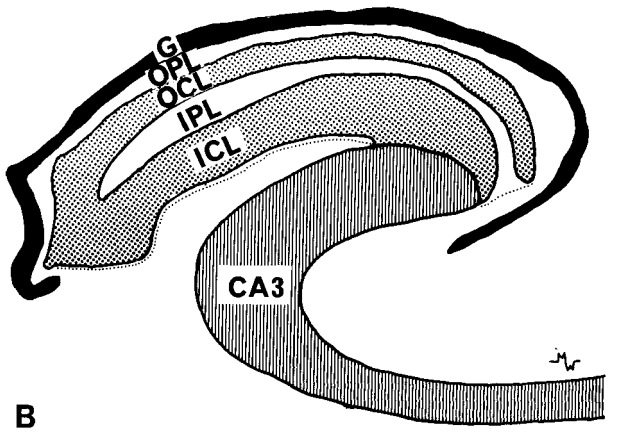
counted. Particles transected by one of the planes, the so-called reference plane, and not by the other, the so-called look-up plane,[2] are defined as particles belonging to the disector volume. The volume defined by the counting frame of known area *a(fra)* positioned on the reference plane times the distance between the two disector planes h is the volume of the disector. Unbiased rules (Gundersen, '77) are used for counting the particles (Q^-) in the frame that are not seen in the other plane. The numerical density (N_V) of the particles is:

$$N_V = \frac{Q^-}{h \cdot a(fra)}$$

In thick transparent sections, it is possible to use the limited focal depth of high numerical aperture lenses to make optical sections at varying depths of the section. The optical disector is an extension of the disector principle (Gundersen, '86, Fig. 2.3; Gundersen et al., '88; Brændgaard et al., '90). It can be thought of as a series of consecutive disectors in which the distance between the first look-up plane and the last reference plane may be arbitrarily great and in

[2]This plane is referred to as the look-up plane because one uses it to "look up" the particles (in this case nuclei) that are present in the reference plane.





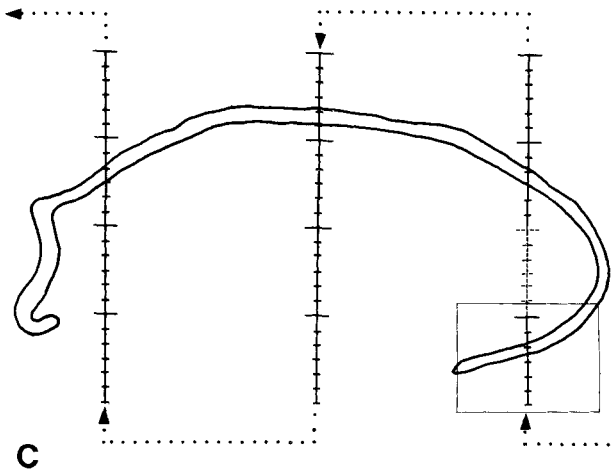


Fig. 7. **A:** Frontal section through the hippocampus at level 7 of Figure 1 showing the classic topological relationship of the subdivisions in a transverse section. **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A. F, fornix.

Fig. 8. **A:** Frontal section corresponding to level 8 of Figure 1. **B:** Diagram of the neuron-containing layers onto which has been superimposed a representation of the point-counting grid used to make the determinations of the areas of the layers. Each + is separated by 1 mm. The circled +s were used in the determination of the area of CA1. When counting, a transparent grid of this type was placed directly onto the microscope slide and viewed with a dissection microscope. See text.

Fig. 9. **A:** Frontal section through the hippocampus at level 9 of Figure 1. **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A. F, fornix.

Fig. 10. **A:** Frontal section through the hippocampus at level 10 of Figure 1. **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A. F, fornix.

Fig. 11. **A:** Frontal section through the hippocampus at level 11 of Figure 1. This level corresponds to the subsplenial gyrus, which passes caudally under the corpus callosum. Sections from this level were not used in the analysis (see text). **B:** Diagram of the neuron-containing layers of the subdivisions in the section shown in A. F, fornix; CC, corpus callosum.

Fig. 12. **A:** Frontal section through the body of the hippocampus showing the cellular organization of the dentate gyrus and part of CA3. **B:** Outlines of the cellular layers of the dentate gyrus shown in A. The stippled line depicts the inner border of the hilus as it was defined for quantitative study. The outer border of the hilus corresponds to the outer border of OCL. G, granule cell layer; OPL, outer plexiform layer; OCL, outer cell layer; IPL, inner plexiform layer; ICL, inner cell layer. **C:** Diagram showing the outline of the granule cell layer shown in A and three rows of points used to estimate the volume of the granule cell layer. The rectangle depicts the field viewed on the video monitor during the counting of points that hit the layer and the size of the six-point array relative to the field. When the field of view is moved up by a distance that is slightly less than the height of the screen, the point array (broken lines) has its lower point at a position that roughly corresponds to the upper point in the previous field. The dotted lines indicate the systematic movements of the field of view that start in a randomly positioned row *outside* the region to the right.

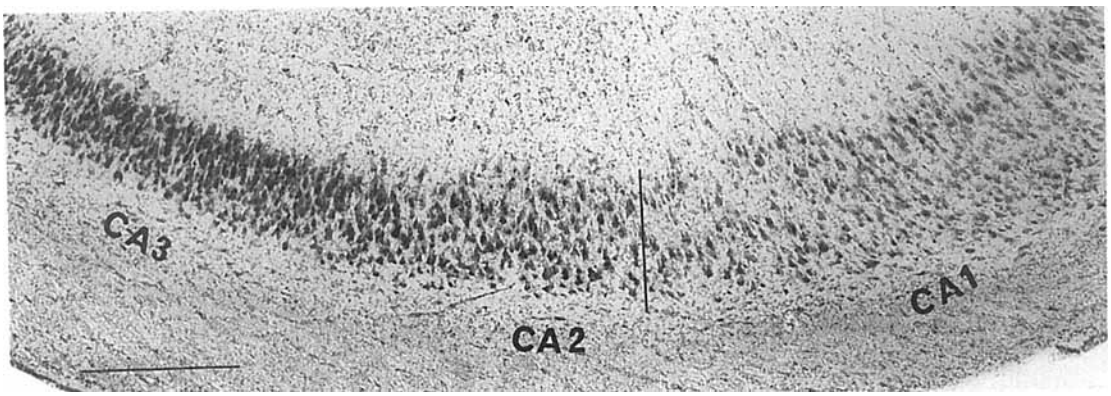


Fig. 13. Micrograph showing the cellular organization of Ammon's horn. The vertical line indicates the border between the two divisions of the pyramidal cell layer that were quantified, i.e., CA3-2 and CA1. Magnification bar = 0.5 mm.

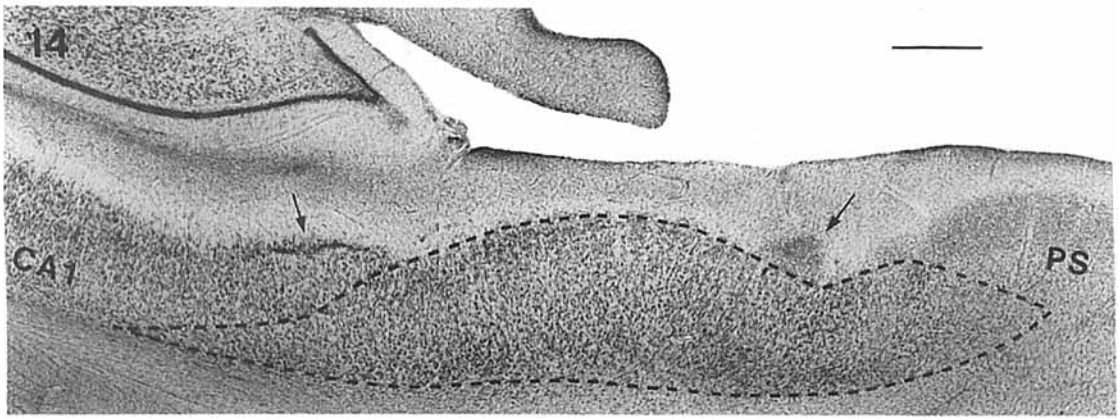
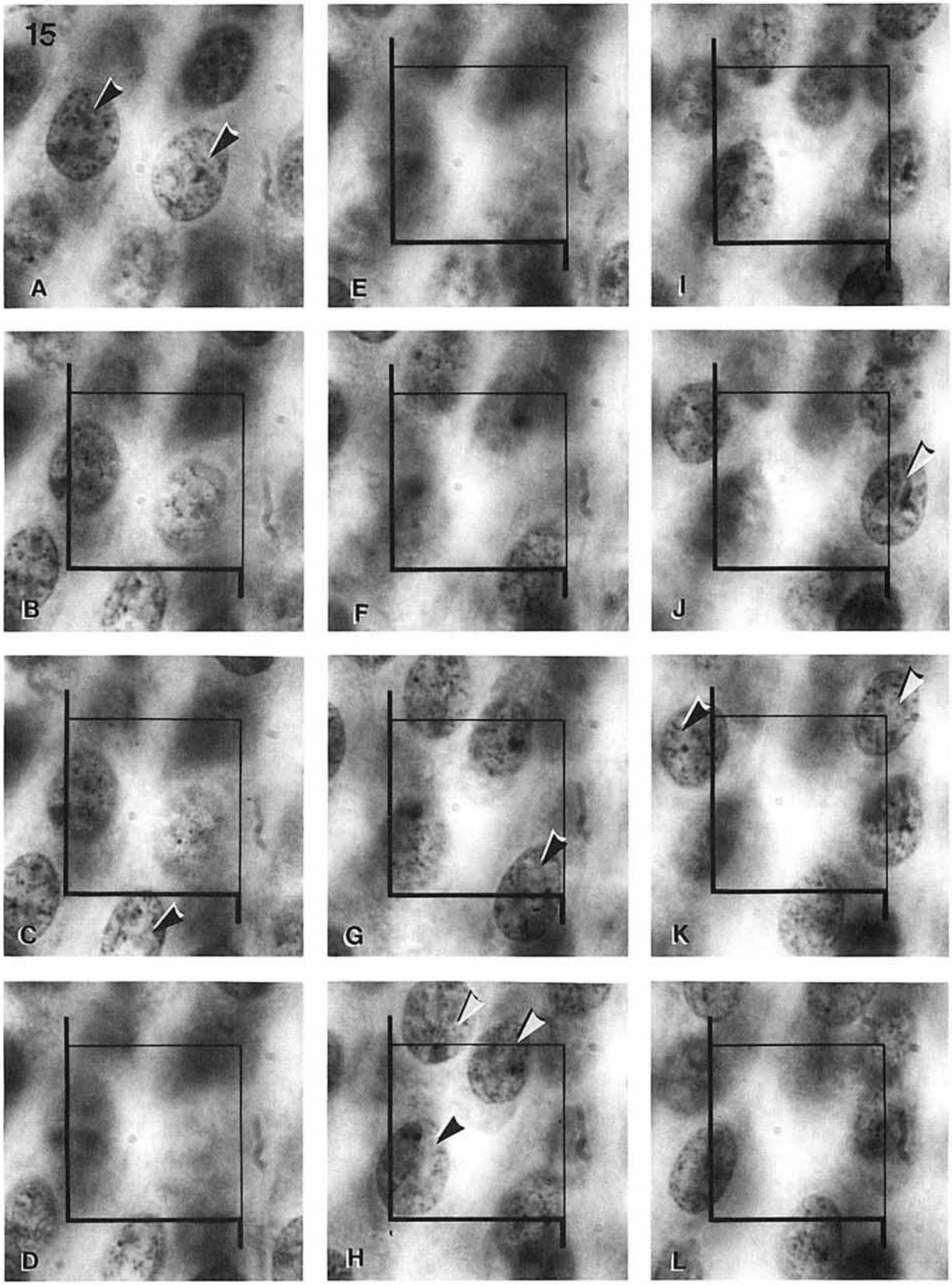


Fig. 14. Outline of the neuron-containing layers of the subiculum as it appears on a transverse section of the hippocampus at level 7 of Figure 1. The left arrow points to a group of small pyramidal cells that could often be observed superficial to the deep lateral border of the subiculum. These cells are defined as part of the prosubiculum by some authors (see text). In sections in which these cells were observed in this study, they were included within the boundaries of the cellular layers of CA1. The right arrow points to an isolated group of presubicular neurons, which were often observed superficial to the cellular layers of the subiculum. These clusters of neurons were always excluded from the boundaries of the subiculum. PS, presubiculum. Magnification bar = 1 mm.

Fig. 15. A series of optical disectors through the granule cell layer of the dentate gyrus used for making an estimate of the numerical density (N_V) of granule cells. An unbiased counting frame of known area (0.02 mm × 0.02 mm) is superimposed on an optical section obtained with a high numerical aperture oil objective. Each optical section shown (**A–L**) is separated by 0.002 mm. Starting at the first look-up section, A, the nuclei that come into focus within the frame are counted, Q^-, as one proceeds to focus through a known distance of the thick section. Profiles of nuclei in contact with the thin edges of the frame are considered to be inside the frame. Those touching the thick forbidden line are defined as being outside the frame. The two nuclei in focus at the look-up level shown in A (black arrows) are not counted because they do not *come into focus* as one proceeds to focus through the sample. The first nucleus to come into focus does so at C (black arrow), but it is not counted because it touches the forbidden line of the counting frame. Another nucleus comes into focus at G, but it is not counted because it also touches the forbidden line. Three nuclei come into focus at H; the two upper nuclei that touch the thin line of the frame (white arrows) are considered to be in the frame and are counted. The nucleus at the lower left of the frame in H (black arrow) is not counted. At J a new nucleus comes into focus (white arrow) at the right of the frame and is counted because part of the profile lies within the frame, and it does not contact the forbidden line. At the last reference section, K, two new nuclei come into focus. The nucleus at the upper right (white arrow) is counted; that at the upper left (black arrow) is not counted. Between levels A and K, a distance of 0.02 mm, four nuclei (white arrows) are counted, i.e., $Q^- = 4$. The volume of the sample is 0.000008 mm^3. The sample N_V is 4/0.000008 mm^3 or 0.5×10^6 mm^{-3}, which is twice the average N_V of the granule cell layer. Note that the division of the total disector height of 0.02 mm into steps of 0.002 mm is arbitrary and only for the purpose of illustration. In practice, one simply focuses down from levels A to K in one slow movement and counts the four nuclei as they come into view. The last section shown, L, is necessary only for resolving ambiguities at the last reference level, K, when the nuclei are nonconvex, an unusual feature in the case of neuronal nuclei.


15
A
E
I
B
F
J
C
G
K
D
H
L

which the optical section functions as both reference and look-up planes as it is moved through the section. In practice, one focuses down through a known thickness of the section h in one continuous movement and counts how many *new* neuronal nuclei *come into focus* within the counting frame (Fig. 15).

The optical disector was preferred to three-dimensional counting rules (Howard et al., '85; Williams and Rakic, '88, '89) because it eliminates the need to follow the edges of the nuclei throughout the three-dimensional test system. With optical disectors, the nuclei are counted when they *first* come into focus. Whether or not the profiles of the counted nuclei fall inside or outside of the test system at other levels is immaterial (see Fig. 2.2 in Gundersen, '86).

The area *a(fra)* of the unbiased counting frame, the height h of the optical disector, and the number of samples used in each subdivision were determined a priori from the following considerations. The expected coefficient of error CE for n independent observations made from a Poisson distribution is $1/\sqrt{n}$. To achieve an estimate of N_V within an individual with a CE less than 0.10, approximately 100 observations would have to be performed in a systematically random manner. Based on a preliminary analysis of N_V in the different subdivisions, the areas of the unbiased counting frames were dimensioned so that one neuronal nucleus would typically be sampled in a disector with a height of 20 μm. This height was chosen because it was a comfortable distance over which to make observations. The dimensions of the counting frame used to obtain this sampling intensity in the granule cell layer (Fig. 15) was 0.02 mm $\times$ 0.02 mm. In hilus (Fig. 16A) and in subiculum (Fig. 16D), it was 0.1 mm $\times$ 0.1 mm, and in CA3-2 (Fig. 16B) and CA1 (Fig. 16C), it was 0.06 mm $\times$ 0.06 mm. It should be emphasized that there is nothing sacred about the absolute value of these parameters. They serve only as a rational starting point and can be adjusted during the study. The important considerations are obtaining the desired CE of the estimate without extra work and insuring that the sampling is done systematically random—that is, with a random start and thereafter with a predetermined regularity.

In practice, the systematic placement of the counting frame on the layer was accomplished by moving the section in a raster pattern with steps of 0.7 mm while observing a video image of the microscope field at a relatively low magnification. The area of the field that corresponded to the field seen with a high-power oil objective was marked. When a sample was to be taken, the oil objective was moved into place and the appropriate counting frame placed in the field. The step movement used in the low-magnification raster scan was dimensioned so that samples were taken at each step in the granule cell layer and in CA3-2, (i.e., each time the marked high-power field fell within the delineations of these layers), at every third step in the hilus, at every sixth step in subiculum, and at every 11th step in CA1. These sampling frequencies provided the necessary number of observations in the different subdivisions when all of the sections were used. The optical disector sampling was performed on video images onto which the counting frame was superimposed with a computer equipped with a video interface. A 100×, N.A. 1.40 oil-immersion objective was used when sampling with the optical disector. When the counting frame did not fall entirely within the layer, the number of frame corners within the layer was noted and the frame recorded as a fractional frame, the fraction being that of the corners of the frame hitting the layer.

RESULTS

The means of the number of neurons N ($OCV = SD/mean$, observed relative variance) in the subdivisions of the five human hippocampi studied were: granule cell layer 15.4×10^6 (0.28), hilus 1.98×10^6 (0.16), CA3-2 2.70×10^6 (0.21), CA1 16.4×10^6 (0.32), and subiculum 4.51×10^6 (0.19). For each subdivision of each hippocampus, the data used to make the estimates of N are presented in Table 2. There was no significant correlation between either the brain weight or the height of the patients and the number of neurons in the various hippocampal subdivisions. There was, however, a significant correlation between the brain weight and height of the patients (Pakkenberg and Voigt, '64).

A significant correlation between neuron number and age was obtained in only one of the subdivisions, CA1. This was an inverse relationship with a coefficient of correlation $r = -0.88$ (Fig. 17).

Analysis of the sampling scheme

To evaluate the appropriateness of the sampling scheme, it is necessary to determine the degrees to which the biological variation (CV) and the stereological sampling variation (OCE) of the estimates, at the level of the individuals, contributed to the observed relative variation (OCV) in neuron number for the five hippocampi. The compound estimator of total neuron number (N) in each subdivision

$$N = N_V \cdot V(ref)$$

may be analyzed in terms of its two components. This can be done only at the level of each individual because the product may show a large or a small variation among individuals, depending on the *correlation* between the absolute volume and the numerical density at the level of individuals. For example, if larger hilar regions have lower numerical densities, and vice verse (as is indeed the case), there would tend to be a relatively small variation in the total number of hilar neurons among individuals. If larger CA1 regions have higher numerical densities than smaller CA1 regions (as is also the case), there would tend to be a relatively large variation in the total number of CA1 neurons among individuals.

When the constants related to the point densities of the counting grids and the absolute volumes of the disectors are ignored, an analysis of ΣP and $\bar{Q}^-$, at the level of an individual, corresponds to an analysis of V and N_V. A detailed example of such an analysis for the CA1 region of the 85-year-old patient is shown in Table 3 and Figure 18. In this example, the $CE(\Sigma P)$ of the estimate of the total volume of the pyramidal cell layer of CA1 is 0.047. (The average value for the CEs of the various subdivisions of the five hippocampi that were studied varied from 0.05 to 0.11. The total number of points counted on the cross-sectional areas varied from 50 to 207 per subdivision.) As shown in Figure 18B, the average number of nuclei counted per frame in a section, $\bar{Q}^- = Q^-/F$, varied somewhat from section to section. This quantity is the ratio of the total number of nuclei counted per section divided by the number of disectors, $\Sigma Q^-/\Sigma F$. The formula for calculating the CE of such a ratio is given in Table 3, Formula 3 (see Gundersen and Jensen, '87, Equation 16). The CEs of the total number of nuclei counted, $CE(\Sigma Q^-)$, and of the number of frames, $CE(\Sigma F)$, are calculated in the same manner as the $CE(\Sigma P)$ shown in Table 3.

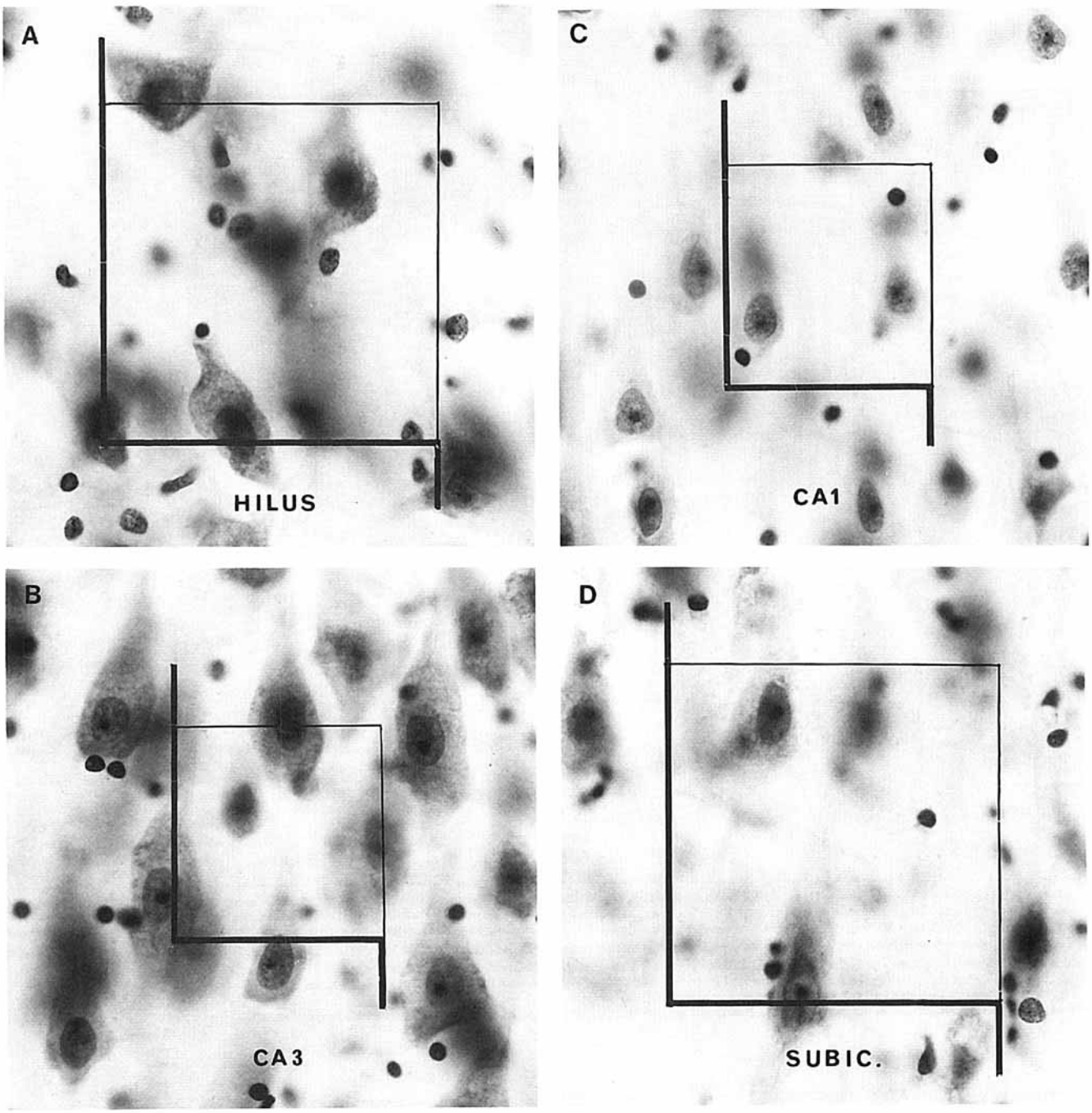


Fig. 16. High-power light micrographs showing the counting frames used with the optical disectors in four subdivisions of the hippocampus. **A:** Hilus, the width of the frame is 0.1 mm; **B:** CA3-2, 0.06 mm; **C:** CA1, 0.06 mm; and **D:** Subiculum, 0.1 mm.

The formula for calculating the covariance of the numerator and denominator under systematic sampling is Formula 2 in Table 3 (Gundersen and Jensen, '87, Equations 14 and 15). In the example (Fig. 18B), the final CE for $\Sigma Q^-/\Sigma F$ (the mean number of neurons per disector), referred to as $CE(Q^-)$, is 0.052. (The $CE(Q^-)$ for the different subdivisions varied from 0.04 to 0.08.) This comfortably low CE of the numerical density estimate was reached by counting on the average just above 100 neurons in ~70 disectors per subdivision. Because the numerical density of neurons within any given subdivision is rather homogeneous, as evidenced by the small $CE(Q^-)$-values, it is easier to design the sampling scheme. Because similar numbers of neurons are counted in each subdivision to produce the desired $CE(Q^-)$s, the relatively large differences in the volumes of the neuron-containing layers of the subdivisions can be dealt with by simply modulating the size of the disector frames and the sampling fraction. Analysis of the two esti-

TABLE 2. Data From the Five Subdivisions of the Five Hippocampi Showing Individual Values and the Means, Standard Deviations, and Coefficients of Variation for the Group[1]

Age (years)	No. of sections[2]	ΣP	$CE(\Sigma P)$	ΣQ^-	No. of frames[3]	$\overline{Q^-}$	$CE(\overline{Q^-})$	V (mm^3)	N_V (10^6 mm^{-3})	N (10^6)	$CE(N)$
Granule cell layer											
47	13	207	0.095	124	62.00	2.00	0.060	77.8	0.250	19.5	0.113
56	10	128	0.069	70	41.75	1.68	0.049	48.1	0.210	10.1	0.057
66	9	143	0.084	121	43.75	2.77	0.068	53.7	0.346	18.6	0.071
80	10	188	0.052	70	54.75	1.28	0.033	70.7	0.160	11.3	0.058
85	11	173	0.063	131	60.25	2.17	0.040	65.0	0.272	17.7	0.063
Mean	10.6	168	0.074	103	52.5	1.98	0.052	63.1	0.247	15.4	0.075
SD	1.5	32		31	9.3	0.56		12.1	0.070	4.4	
CV	0.064	0.192		0.296	0.178	0.281		0.192	0.281	0.284	
									Biol =	0.274	
Hilus											
47	13	81	0.068	86	57	1.51	0.048	243	0.00754	1.83	0.095
56	10	50	0.102	132	58	2.28	0.071	150	0.01138	1.71	0.158
66	9	73	0.111	99	54	1.83	0.076	219	0.00917	2.01	0.163
80	11	89	0.078	84	61	1.38	0.111	267	0.00689	1.84	0.060
85	11	99	0.067	115	68	1.69	0.055	297	0.00846	2.51	0.030
Mean	10.8	78.4	0.087	103	59.6	1.74	0.076	235	0.00869	1.98	0.114
SD	1.5	18.6		20	5.3	0.35		56	0.00174	0.32	
CV	0.061	0.237		0.197	0.089	0.200		0.237	0.200	0.160	
									Biol =	0.112	
CA3-2											
47	12	51	0.091	109	83.00	1.31	0.037	153	0.0182	2.79	0.118
56	9	51	0.131	326	82.75	3.94	0.028	153	0.0197	3.01	0.124
66	9	42	0.117	122	95.00	1.28	0.071	126	0.0178	2.25	0.107
80	10	47	0.113	97	94.75	1.02	0.013	141	0.0142	2.00	0.115
85	11	68	0.097	141	116.00	1.22	0.024	204	0.0169	3.44	0.100
Mean	10.2	52	0.111	159	94.3	1.76	0.040	155	0.0174	2.70	0.113
SD	1.3	10		95	13.5	1.23		29	0.0020	0.58	
CV	0.057	0.189		0.596	0.144	0.699		0.189	0.117	0.215	
									Biol =	0.183	
CA1											
47	15	93	0.052	103	66.5	1.55	0.075	1120	0.0215	24.0	0.091
56	13	86	0.053	190	62.5	3.04	0.034	1030	0.0152	15.7	0.048
66	12	85	0.048	78	58.0	1.35	0.034	1020	0.0187	19.1	0.075
80	12	85	0.060	51	61.0	0.84	0.066	1020	0.0116	11.8	0.085
85	15	81	0.047	55	65.0	0.85	0.052	970	0.0118	11.4	0.080
Mean	13.4	86	0.052	95	62.6	1.52	0.055	1032	0.0158	16.4	0.077
SD	1.5	4		57	3.3	0.90		52	0.0043	5.3	
CV	0.051	0.051		0.596	0.053	0.593		0.051	0.275	0.321	
									Biol =	0.312	
Subiculum											
47	14	200	0.046	104	69.75	1.49	0.062	600	0.00746	4.47	0.061
56	12	143	0.062	117	46.00	2.54	0.050	429	0.01272	5.46	0.079
66	12	190	0.032	95	56.00	1.70	0.038	570	0.00848	4.83	0.058
80	12	185	0.056	86	50.75	1.70	0.048	555	0.00847	4.70	0.087
85	14	128	0.039	77	48.00	1.60	0.042	384	0.00802	3.08	0.074
Mean	12.8	169.2	0.048	96	54.1	1.81	0.049	508	0.00903	4.51	0.073
SD	1.1	31.7		16	9.5	0.42		95	0.00210	0.88	
CV	0.038	0.187		0.162	0.176	0.233		0.187	0.233	0.195	
									Biol =	0.181	

[1] *SD* = standard deviations; *CV* = coefficients of variation *SD/mean*. Because these are *observed* quantities, they are noted as *OSD* and *OCV* in the sections of the text dealing with the distribution of variances. ΣP = The total number of points counted over the sectional profiles of the neuron-containing layer(s) of a subdivision (to be multiplied by a constant to obtain *V*). $CE(\Sigma P)$ = The observed variance of the estimate ΣP, which also expresses the variance of the estimate of *V*. An example of how this is calculated is given in Table 3. ΣQ^- = The total number of neuronal nuclei counted on all sections with optical disectors. $\overline{Q^-}$ = The mean number of neuronal nuclei per frame (i.e., per disector sample). $CE(\overline{Q^-})$ = The variance of $\overline{Q^-}$, the calculation of which is presented in Table 3. *V* = Volume of the layer of the subdivision in which the estimate of N_V was made. *N* = number of neuronal nuclei per mm^3; $/N$ = total number of neurons ($N = V \times N_V$); $CE(N)$ = computable variation of the stereological estimate of *N*, described in the Discussion; Biol = estimate of the true biological variance of *N*, as described in the text.

[2] The number of sections separated by 3 mm used to make estimates of both N_V and *V*.

[3] The total number of disector samples made on all sections. Noninteger values reflect sampling frames not entirely within the delineated region (see text).

mates, *V(ref)* and N_V, indicates that they contribute roughly equally to the sampling variance at the level of sections; both have *CE*s in the order of 7%.

The final estimate of the total number of neurons (*N*) in a subdivision is

$$N = c \cdot \sum P \cdot \sum Q^- / \sum F$$

where *c* is a constant related to sampling intensities, point density, frame size, and disector height, which may be ignored here because it has no bearing on the distribution of the variance. Note that the above is the unbiased estimator used everywhere in this report and that it is in general different from the estimator shown in the last column of Table 3, $\Sigma N_i = \Sigma(P_i \cdot Q_i/F_i)$. Any one of the three factors, *P*, Q^-, and *F* may be zero for a particular section, with the consequence that there can be a small bias in the estimator of ΣN_i (but not in the above $\Sigma P \cdot \Sigma Q^- / \Sigma F$). The estimator of ΣN_i is extremely useful, however, because it provides a straightforward estimate of the *precision* of the estimate of total neuron number, $CE(N)$. The constant c times the N_i in each row of the estimator of ΣN_i is the total number of neurons in each 3-mm-thick slab. The systematic variation in N_i is the main determinant of $CE(N)$. Treating the N_i values for the slabs just like any other set of systematic data, (i.e., like ΣP and ΣQ^- with Formula 1 in Table 3), we obtain a $CE(N)$ of 0.08 in the example (Table 3, right column). For the various subdivisions, the precision of the estimate of total number varied from 0.07 to 0.11 and is therefore rather close to the precision at the level of the individual specified in the original design of the sampling scheme.

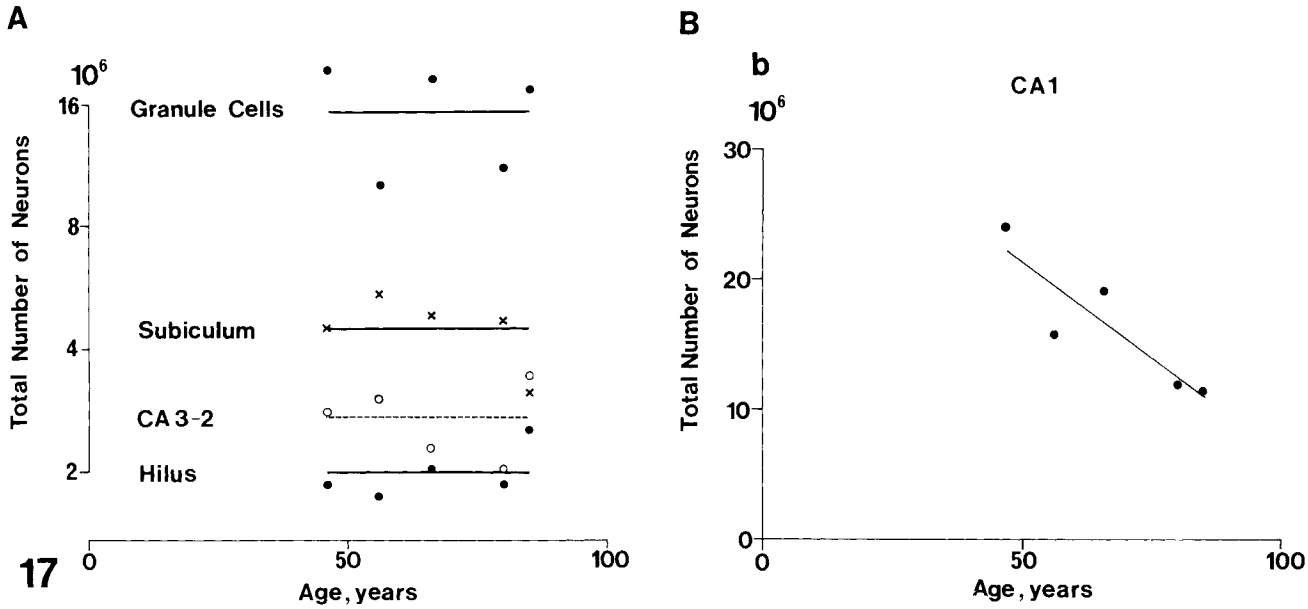


Fig. 17. **A:** The total number of granule cells and neurons in subiculum, CA3-2, and hilus plotted as a function of age. There is no significant correlation between neuron number and age in these four subdivisions. (Note that the numbers of neurons are presented on a logarithmic scale in order to accomodate the data from all four subdivisions on the same graph.) **B:** The total number of neurons in CA1, of the five patients used in this study plotted as a function of age. The regression line is shown. There is a significant negative relationship ($r = -0.88$) between age and neuron number.

As pointed out at the beginning of this report, whether or not this precision is necessary depends on the biological variation, which was unknown until the analysis of the first five human hippocampi had been performed. As shown in Table 2, the observed variation *OCV* among humans for the number of neurons in the various subdivisions varied from 0.16 in the hilus to 0.32 in CA1. In general, the *OCV*s are well above the level of ~0.15 that was used as a basis for the design of the sampling scheme. As illustrated in Figure 17B, a large part of the large variation in the total number of neurons in CA1 can be attributed to age, a phenomenon irrelevant to the design and evaluation of the sampling scheme, however. The residual variation of N(CA1), the scatter with respect to the regression line in Figure 17B, is $2.9 \cdot 10^6$ as opposed to the *SD* of $5.37 \cdot 10^6$. As a consequence, the corresponding relative residual variation, $CRV(N)$, is 0.18 and comparable to that of the other subdivisions.

Table 2 also shows how much of the observed variance can be attributed to real variation among the hippocampi, i.e., to the biological variation (noted in the table as Biol.). The biological variation is easily estimated by subtracting the estimated stereological sampling variance of the individual estimate of total neuron number from the total observed variance (Kroustrup and Gundersen, '83).

$$CV^2(N) = OCV^2(N) - OCE^2(N)$$

The biological variance contributed 93%, 86%, 81%, 72%, and 49% to the observed variance of total neuron number among the five hippocampal subdivisions. On average, three fourths of the total variance was related to real biological differences. This is satisfactory in the sense that the sampling variance introduced by the stereological estimation procedure is a minor fraction of the observed variance; as a consequence, it is not necessary to analyze the sampling scheme further.

Optimizing the sampling scheme

The sampling scheme and the various sampling densities used is this study were selected on the basis of the general principles outlined in General Methods and the data from a pilot study made on two hippocampi. The data obtained through the application of this sampling scheme to five human hippocampi and the detailed analysis of the sampling variances, at the first two sampling levels described above, permits a number of small final adjustments to be made in the procedure. The hilus and pyramidal cell layer of CA3-2 were hit by only 80 and 50 points, respectively, and their shapes make the Cavalieri estimator of total volume relatively imprecise. The use of a separate test system with the points in a 0.9 by 0.9 mm array will improve the sampling slightly without any appreciable increase in the effort. The number of points counted over the relatively voluminous subiculum was too large, however, and it is therefore desireable to change the scheme to one in which the volumes of pyramidal cell layer of CA1, and the cellular layers of subiculum are estimated using a test system with points in a 1.8 by 1.8 mm array, i.e. roughly a twofold lower density in subiculum and a slightly larger density in CA1. For the estimate of sectional areas of the granular cell layer, the density of points can be reduced to ~50%, e.g., an increase in the distance between the rows of 25% to 1.66 mm and an increase in the distance between points in the arrays of ~25% to 0.125 mm. The latter will necessitate a change in the number of points per array because the array is dimensioned to span the height of the video monitor. These changes will distribute the relative point densities slightly

TABLE 3. An Example of How $CE(\Sigma Q^-/\Sigma F)$ and $CE(N)$ Can Be Calculated for One of the Subdivisions of an Individual Hippocampus[1]

Section no.	P_i	$P_i \cdot P_i$	$P_i \cdot P_{i+1}$	$P_i \cdot P_{i+2}$	Q_i	F_i	$Q_i^- \cdot F_i$	$(F_i \cdot Q_{i+1}^- + F_{i+1} \cdot Q_i^-)/2$	$(F_i \cdot Q_{i+2}^- + F_{i+2} \cdot Q_i^-)/2$	$N_i = \frac{Q_i^-}{F_i} \cdot P_i$
1	11	121	143	121	5	9	45	61	63	6.11
2	13	169	143	52	8	10	80	81	26	10.40
3	11	121	44	44	9	9	81	27	36	11.00
4	4	16	16	12	2	4	8	12	9	2.00
5	4	16	12	8	4	4	16	12	6	4.00
6	3	9	6	15	3	3	9	4.5	12	3.00
7	2	4	10	8	1	2	2	6	3	1.00
8	5	25	20	15	4	4	16	10	8	5.00
9	4	16	12	16	1	4	4	3.5	3.5	1.00
10	3	9	12	15	1	3	3	3	7	1.00
11	4	16	20	20	1	3	3	7	9.5	1.33
12	5	25	25	25	4	2	8	13	12	10.00
13	5	25	25	10	5	4	20	19.5	4.5	6.25
14	5	25	10		6	3	18	4.5		10.00
15	2	4			1	1	1			2.00
Σ	81	$A = 601$	$B = 498$	$C = 361$	55	65	$D = 314$	$E = 264$	$G = 199.5$	74.1
CE	0.047				0.067	0.047				0.080

Formula 1:

$$CE(\Sigma P) = \frac{\sqrt{(3A + C - 4B)/12}}{\Sigma P} = 0.047$$

Formula 2:

$$Cov(\Sigma Q^-, \Sigma F) = (3D + G - 4E)/12 = 7.125$$

Formula 3:

$$CE\left(\frac{\Sigma Q}{\Sigma F}\right) = \sqrt{CE^2(\Sigma Q^-) + CE^2(\Sigma F) - \frac{2 \cdot Cov(\Sigma Q^-, \Sigma F)}{\Sigma Q^- \cdot \Sigma F}}$$

$$= \sqrt{0.067^2 + 0.047^2 - \frac{2 \cdot 7.125}{55 \cdot 65}}$$

$$= 0.052$$

[1]The data is from CA1 of the 85-year-old patient shown in Table 2. The $CE(\Sigma P)$ is calculated from Formula 1, which contains three variables A, B, and C, each of which can be calculated from the number of grid points (P_i) counted on each section. The variable A is the sum of the squares of the number of points counted on each section ($P_i \cdot P_i$). The variable B is the sum of the product of the number of points counted on each section and the number of points counted on the next section in the series ($P_i \cdot P_{i-1}$). The variable C is the sum of the products of the number of points on each section and that counted on the second next section in the series ($P_i \cdot P_{i+2}$). In this example $CE(\Sigma P) = 0.047$.
The $CE(\Sigma Q^-/\Sigma F)$ is calculated from Formula 3, which contains three terms. The terms $CE(\Sigma Q^-)$ and $CE(\Sigma F)$ can be calculated from the number of neurons counted in disector samples from each section Q^-, and the number of disector samples from each section, F_i, in the same manner as $CE(\Sigma P)$, described above. In this example, $CE(\Sigma Q^-) = 0.067$ and $CE(\Sigma F) = 0.047$. The third term, the covariance of the numerator and denominator of the ratio, $Cov(\Sigma Q^-, \Sigma F)$ is calculated by using Formula 2 from three variables, D, E, G, each of which can be calculated from Q_i^- and F_i. The variable D is the sum of the products of the number of neurons counted on a section and the number of disector samples used to obtain the counts, $Q_i \cdot F_i$. The variable E is the sum of the means of the product of F from each section and Q^- from the next section in the series plus the product of Q^- from the section and F from the next section in the series, that is $(F_i \cdot Q_{i-1} + F_{i-1} \cdot Q_i)/2$. The variable G is calculated in a similar manner except that the combinations are made between F and Q^- on section pairs separated by one section, that is $(F_i \cdot F_{i+2} + F_{i+2} \cdot Q_i^-)/2$. The covariance $Cov(\Sigma Q, \Sigma F)$ in this example is 7.125. The $CE(\Sigma N)$ shown at the left is calculated in the same manner as $C\Sigma(\Sigma P)$.
The points to be checked with this analysis are 1) that the $CE(N)$ is below the value set as a goal when the sampling scheme was designed (in this case 0.10) and 2) that the $CE(\Sigma P)$ and $CE(\Sigma Q^-/\Sigma F)$ are similar in magnitude, indicating that the variances of $V(ref)$ and N_V, of which they are expressions, respectively, make roughly equal contributions to the variance of N. If the average $CE(N)$ over individuals is too high, more sampling will have to be done. If the contributions to the variance of the estimate of N are markedly unequal, the sampling scheme may have to be adjusted.
How the analysis of variance at the level of the individual is used in the optimization of the sampling scheme is described in detail in Results under "Optimizing the sample scheme."

more evenly among the subdivisions, adjust the precisions, and reduce the effort marginally.

Most of the effort in the estimation procedure is spent in the estimation of the numerical density. In all cases, ~100 neurons were counted in ~70 disectors, and the $CE(N_V)$ was ~10%. Considering the small contribution to the overall variance made by the stereological sampling, this is clearly too precise, and it would be more efficient to increase the distance between disectors by 15 to 20%, that is, increase the distance between the frames in the raster pattern, from which the samples would be drawn in a manner described in Methods to 0.84 mm and thereby reduce the counting effort by 30 to 40%.

Summary of the optimized sampling scheme

1. Frontal sections are collected at 3-mm intervals along the entire length of the hippocampus.
2. The areas of the profiles of the neuron-containing layers of the various subdivisions are estimated by point counting using grids with the following point densities:

a. Hilus	1 point/0.81 mm^2
b. CA3-2	1 point/0.81 mm^2
c. CA1	1 point/3.24 mm^2
d. Subiculum	1 point/3.24 mm^2
e. Granule cell layer	1 point/0.2075 mm^2:
	0.125 mm between the points in an array
	1.66 mm between the rows of arrays

3. The counting frames used with the optical disectors have the following dimensions in the various subdivisions:

a. Granule cell layer	0.02 mm × 0.02 mm = 0.0004 mm^2
b. Hilus	0.1 mm × 0.1 mm = 0.01 mm^2
c. CA3-2	0.06 mm × 0.06 mm = 0.0036 mm^2
d. CA1	0.06 mm × 0.06 mm = 0.0036 mm^2
e. Subiculum	0.1 mm × 0.1 mm = 0.01 mm^2

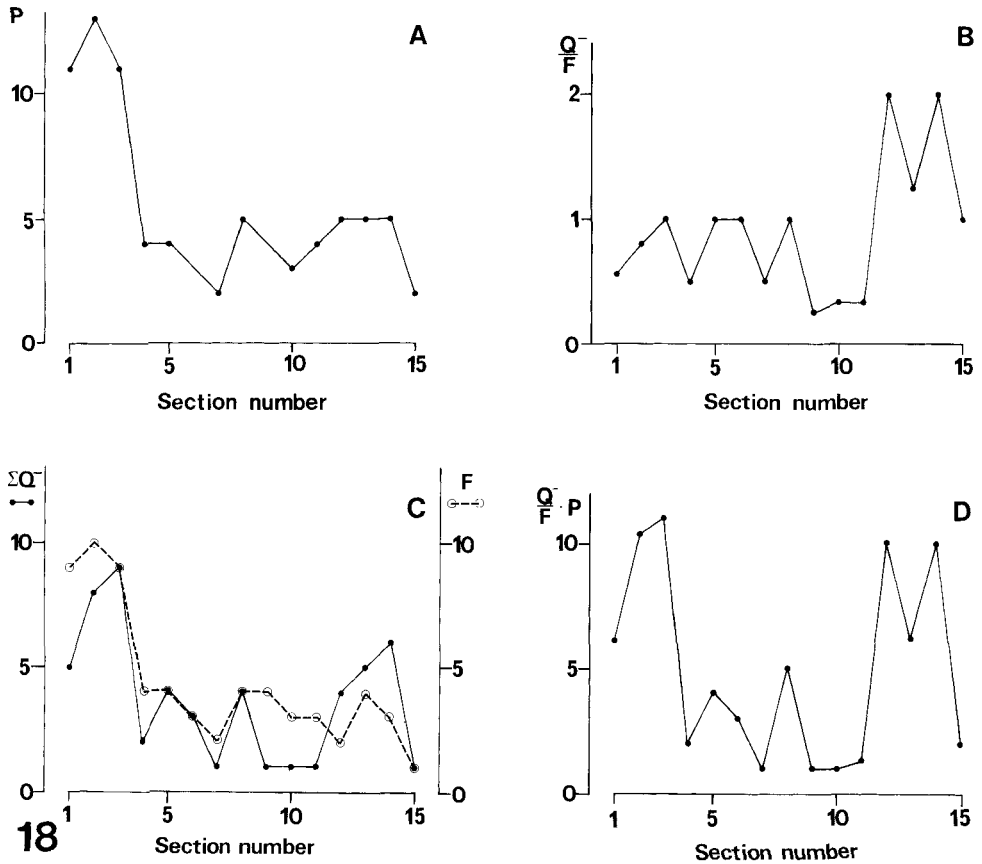


Fig. 18. Graphic representations of the data presented in Table 3. The data are drawn from the CA1 region of the 85-year-old patient used in this study. **A:** The number of points (P) in the counting grid hitting the CA1 region on each of the 15 sections used in the analysis is plotted as a function of the position of the section in the series. This corresponds to the data shown in the first column of Table 3, P_i. **B:** Plot of the average number of nuclei (Q^-) per disector sample (F), Q^-/F for each section as a function of the position of the section in the series. **C:** Dual plot of the sum of the nuclei counted in disector samples, ΣQ^- (solid line) and the number of disector samples, F (broken line) for each section as functions of the position of the section in the series. **D:** Plot of the product of the number of points, P, and the number of nuclei per disector sample Q^-/F for each section, as a function of the position of the section in the series. This corresponds to the product of $V_{(ref)}$ and N_V; that is, N, stripped of constants related to the absolute volumes of the points (P) and the disectors (F). The relatively large numbers of neurons present in the first and last sections of the series are related to the large areas of the first sections (A) and the larger numerical densities in the last sections (B), respectively.

4. The height of all optical disectors is 0.02 mm.
5. The raster of frames from which the optical disectors are drawn from each section have an interframe distance of 0.84 by 0.84 mm. The frames are sampled with the following frequencies in the various subdivisions:

a. Granule cell layer	All frames
b. Hilus	Every 3rd frame
c. CA3-2	All frames
d. CA1	Every 11th frame
e. Subiculum	Every 6th frame

Effects of shrinkage

The shrinkage (or expansion) of the tissue that takes place during the agonal and postmorten phases and the subsequent fixation period prior to cutting the 3-mm-thick slabs does not influence the estimate of N. It is important, however, to determine if there is any distortion of the tissue during the further processing of the slabs, that is, from the point at which the slabs are cut to the point at which the sectional areas and numerical densities are estimated. To evaluate changes that take place during these phases of the procedure, the areas of the cut surfaces of nine slabs from three different hippocampi were estimated by point counting under a dissection microscope. A transparent lattice of points, with an interpoint distance of 1 mm, was placed directly onto the cut surface of the freshly cut slabs, and the number of points hitting the tissue was used for estimating the area of the surface of the slab. The slabs were embedded, and one 70-μm section from each block cut, mounted, stained, and covered. The sum of the points for the three slabs and the three corresponding sections for each of the hippocampi are shown in Table 4.

The conclusion that can be drawn from these measurements is that there is a tendency for the area of a section, and therefore the profile area of the layers of a hippocampal subdivision, to enlarge during tissue processing. Although the difference in the areas was not significant in a strictly statistical sense, the sample is small, and it would seem prudent to continue monitoring changes of this type in future studies and experiment with modifications in the procedure for the collection of sections according to the suggestions of

TABLE 4. Point-Counting Data Used to Evaluate Changes in the Dimensions of the Tissue During Processing[1]

	ΣP(3 slabs)	ΣP(3 sections)	Ratio
Hippocampus 1	650	691	1.06
Hippocampus 2	610	617	1.01
Hippocampus 3	624	690	1.11
			Mean = 1.058
			SD = 0.047

[1]Left column, the sum of points hitting the cut surface of three 3-mm-thick slabs taken arbitrarily from three different hippocampi. In the center column, the sum of points hitting sections cut from the three embedded slabs from each hippocampus after the sections were stained and mounted. The ratio of the sum of points counted over the slabs to that counted over sections, indicating a slight expansion of the tissue, is shown on the right along with the mean and *SD* of the ratio.

Gerrits et al. ('87). No corrections for alterations in tissue volume during tissue processing were made in the data presented here.

Reproducibility of the delineations of the neuron-containing layers of the subdivisions

The ability to define consistently the borders of the cellular layers of the hippocampal subdivisions affects the variance of the estimates of *V(ref)* and, as a consequence, affects the variance of the estimates of *N*. To evaluate this effect, the areas of the neuron-containing layers of the five subdivisions were reestimated on all of the sections of two hippocampi, after the layers were redefined without reference to the original boundaries, i.e., after removal of the delineations. The data from the two estimates are shown in Table 5 in terms of the number of points counted with the grids specified in Materials.

Because the *CE* is of the same magnitude as the observed relative variance of the estimate of V, which can be attributed to the point counting technique (*CE*(ΣP), Table 2), it can be concluded that only a small fraction of the variance of the volume estimate can be attributed to variation in the delineations of the neuron-containing layers.

DISCUSSION

Impact of unbiased estimates of the total number of neurons

The technique for estimating the total number of neurons in a defined region described here is free of assumptions about the size, shape, orientation, and distribution of neurons, is readily performed, and is intuitively easy to understand. Because the estimates that it yields are unbiased (i.e., approach the true value as the sample size increases), they can be expected to be highly reproducable in different settings. The widespread application of these principles can contribute to elimination of the controversy about age-related neuron loss (Coleman and Flood, '87), satisfy the call for standardized morphometric techniques (Beach and McGeer, '87; Brizzee, '87; Iversen, '87; Scheff, '87), and convince the critics of the morphometric approach to the study of aging (Bondareff, '87; Saper, '87), or any other process in which the quantitation of the structural dynamics may be of interest, that the new stereological tools can make a contribution.

TABLE 5. Point-Counting Data Used to Evaluate the Effect of the Delineation Process on the Variability of the Estimate of *V(ref)*[1]

	Gran.	Hilus	CA3-2	CA1	Subic	Sum
Hip 1	153	109	49	86	150	547
Repeat	170	105	62	92	139	568
Difference	−17	4	−13	−6	11	−21
Hip 2	178	75	39	84	144	520
Repeat	167	76	48	84	154	529
Difference	11	−1	−9	0	−10	9

[1]Mean *SD* on a single delineation of a single subdivision = $\sqrt{\Sigma diff^2}/2 = 6.72$ points. The observed overall average *CE* of a delineation of a single subdivision is 6.72/mean number of points per subdivision = 0.062. Two independent delineations of the neuron-containing layers of the various hippocampal subdivisions were made on the same series of sections. The point-counting grids were the same as those used in the quantitative analysis. The average *CE* of a single delineation is similar to that of the individual volumes of the layers, indicating that variations in the delineations contribute little to the estimator of *V(ref)*.

Application of these principles to a wider range of material can be expected to provide a starting point for the identification of the anatomical correlates of the cognitive decline associated with aging. The estimation of total neuron number should be considered to be only a starting point because other factors, such as synapse number, transmitter levels, receptor status, dendrite length, and perikarial size, must ultimately be drawn into a comprehensive discussion of performance. Before changes in these parameters can be interpreted in a meaningful manner, however, they will, in most cases, have to be related to the total number of neurons in the area being studied. Although the details of an optimal sampling scheme (i.e., size of the sample frame, distance between samples) for other regions of the nervous system can be expected to be different from those used here, due to differences in volume and neuron density, the basic approach used here, involving the systematic random sampling of a total of about 100 neurons on ten sections, should provide efficient estimates of total neuron number in virtually all parts of the central nervous system.

Within the human hippocampus, one of the most promising applications is the quantitative examination of the relationship of changes associated with normal aging to the histopathological lesions associated with Alzheimer's disease. There appears to be a predilection for specific subdivisions to develop plaques, neurofibrillary tangles, and to lose neurons and, unlike other parts of the brain, there is little overlap of the quantitative estimators of this pathology in normal subjects and patients afflicted with the disease (Ball, '78). A more rigorous unbiased estimation of the total numbers of plaques, neurons characterized by neurofibrillary and granulovacuolar degeneration, and normal neurons can establish the degree to which these phenomena are temporally linked and contribute to an understanding of the focality of these lesions and the identification of the mechanisms by which they are linked.

Loss of CA1 neurons with age

The initial intent of this study was to demonstrate how the application of recently developed stereological techniques can be used to estimate the total number of neurons in the subdivisions of the human hippocampus and to describe an efficient sampling scheme for making the estimates. As described in detail, the main consideration when designing such a scheme is identification of the variation among individuals. To design an optimal scheme for studying the effects of aging, material was collected from a representative group of individuals who ranged from 47 to 85 years of age. Upon completion of the analysis, a significant inverse correlation was observed between age and neuron number in CA1. In none of the other subdivisions was there a correlation between age and neuron number (Fig. 17). In spite of evidence from studies of numerical densities performed in numerous species, including man, that supports this finding (Flood and Coleman, '88), the small num-

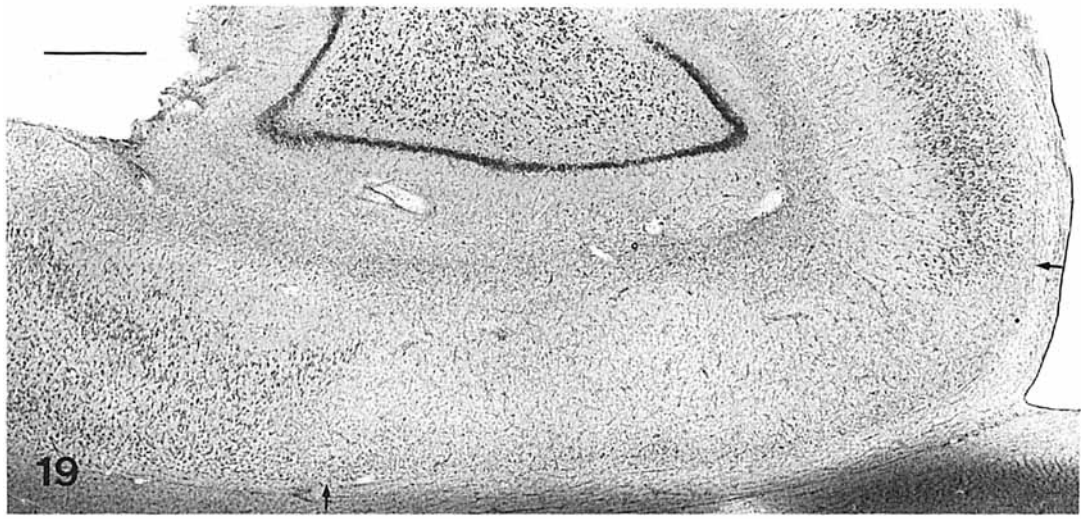


Fig. 19. A transverse section through the hippocampus of an 85-year-old patient showing a localized region of neuron loss in CA1 (between arrows). The section was taken from level 8, Figure 1.

ber of hippocampi included in the present study obliges one to be cautious in drawing rigorous conclusions from the data presented here. The observed decrease in CA1 neurons with age clearly needs to be confirmed in a larger study. It is clear, however, from the work of Zola-Morgan et al. ('86) that the selective loss of neurons in CA1 can account for a large part of the memory deficit associated with aging and age-related dementias.

It may be of value to point out that the lower number of neurons in CA1 in the older brains studied here is a consequence of localized cell loss, rather than a general cell loss uniformly distributed throughout the entire region. In the CA1 region of the brains from the 80- and 85-year-old patients, neuron-free regions having a width and breadth of 1–2 mm were encountered in the caudal parts of CA1 (Fig. 19). The CA1 region is the hippocampal subdivision in which the cytoarchitectural organization and size have changed most during mammalian evolution (Stephan, '83; West, '90), and its structural integrity has been reported to be particularly susceptible to ischemia (Zola-Morgan et al., '86) and epileptic activity (Sommer, 1880; Dam, '80). It has also been suggested by Maragos et al. ('87) that glutamate dysfunction in CA1 may be involved in the selective vulnerability of this part of the hippocampus to pathological processes associated with Alzheimer's disease. Because of the limited scope of the present paper, the relationship between these unique features of CA1 and the selective loss of neurons with age reported here is only speculative and must remain the topic of more comprehensive studies.

In addition to providing an opportunity for drawing attention to the selective loss of CA1 neurons with age, this somewhat serendipitous finding prompts one additional comment related to the value of estimates of total neuron number. Despite the limited size of the sample used in this study, the data presented here constitute the first direct evidence of a decrease in the total number of CA1 neurons with age. The astute, although still unenlightened, reader may argue that this loss is reflected in the numerical densities of neurons and that one might just as well use neuron density (Table 2, CA1, N_V) as a measure of neuron loss. It is important to realize that one does not know that this is true before estimating the total number of neurons. One is again referred to the work of Swaab and Uylings ('87) and Pakkenberg and Gundersen ('88) for examples of situations in which neuron density does not reflect total neuron number.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the assistance of Maj-Britt Lundorf with the histology, Albert Meier with the photography, Anette Funder with the graphic work, and Pernille Schirmer with the typing. The authors also thank Bente Pakkenberg and Arne Møller of the Neurological Research Laboratory, Hvidovre Hospital, Denmark, for their assistance in obtaining part of the hippocampal material used in this study. This work was supported by the Danish Medical Research Council, Fonden til Forskning af Sindslidelser, and BICO, Copenhagen.

LITERATURE CITED

Abercrombie, M. (1946) Estimation of nuclear populations from microtome sections. Anat. Rec. *94*:239–247.

Ball, M.J. (1978) Topographic distribution of neurofibrillary tangles and granulovacuolar degeneration in hippocampal cortex of aging and demented patients. A Quantitative study. Acta Neuropathol. (Berl.) *42*:73–80.

Beach, T.G., and E.G. McGeer (1987) Further methodological problems in cell counting. Neurobiol. Aging *8*:570–571.

Bondareff, W. (1987) Changes in the brain in aging and Alzheimer's disease assessed by neuronal counts. Neurobiol. Aging *8*:562–563.

Braak, H. (1972) Zur pigmentarchitektonik der Grosshirnrinde des Menschen. II. Subiculum. Z. Zellforsch. *131*:235–254.

Braak, H. (1980) Architectonics of the Human Telencephalic Cortex. Berlin: Springer.

Brændgaard, H., and H.J.G. Gundersen (1986) The impact of recent stereological advances on quantitative studies of the nervous system. J. Neurosci. Methods *18*:39–78.

Brændgaard, H., S.M. Evans, C.V. Howard, and H.J.G. Gundersen (1990) Total number of neurons in human neocortex unbiasedly estimated using optical disectors. J. Microsc. *157*:285–304.

Brizzee, K.R. (1987) Neuron numbers and dendritic extent in normal aging and Alzheimer's disease. Neurobiol. Aging *8*:579–580.

Cavalieri, B. (1966) Geometria Degli Indivisibili. Torino: Unione Tipografico, Editrice.

Coleman, P.D., and D.G. Flood (1987) Neuron numbers and dendritic extent in normal aging and Alzheimer's disease. Neurobiol. Aging *8*:521–545.

Curcio, C.A. and K.R. Sloan, Jr. (1986) Computer-assisted morphometry using video-mixed microscopic images and computer graphics. Anat. Rec. *214*:329–337.

Dam, A.M. (1980) Epilepsy and neuron loss in the hippocampus. Epilepsia *21*:617–629.

Duvernoy, H.M. (1988) The Human Hippocampus. An Atlas of Applied Anatomy. Munich: J.F. Bergmann.

Floderus, S. (1944) Untersuchungen über den Bau der menschlichen Hypophyse mit besonderer Berücksichtigung der quantitativen mikromorphologischen Verhältnisse. Acta Pathol. Microbiol. Immunol. Scand. *53*:1–276.

Flood, D.G., and P.D. Coleman (1988) Neuron numbers and sizes in aging brain: Comparisons of human, monkey, and rodent data. Neurobiol. Aging *9*:453–463.

Geneser, F.A. (1987) Distribution of acetylcholinesterase in the hippocampal region of the rabbit. J. Comp. Neurol. *262*:594–606.

Gerrits, P.O., M.B.M. van Leeuwen, M.E. Boon, and L.P. Kok (1987) Floating on a water bath and mounting glycol methacrylate and hydroxypropyl methacrylate sections influence final dimensions. J. Microsc. *145*:107–113.

Gertz, S.D., R. Lindenberg, and G.W. Piavis (1972) Structural variations in the rostral human hippocampus. Johns Hopkins Med. J. *130*:367–376.

Gundersen, H.J.G. (1977) Notes on the estimation of the numerical density of arbitrary profiles: The edge effect. J. Microsc. *111*:219–223.

Gundersen, H.J.G. (1986) Stereology of arbitrary particles. J. Microsc. *143*:3–45.

Gundersen, H.J.G., and E.B. Jensen (1987) The efficiency of systematic sampling in stereology and its prediction. J. Microsc. *147*:229–263.

Gundersen, H.J.G., P. Bagger, T.F. Bendtsen, S.M. Evans, L. Korbo, N. Marcussen, A. Møller, K. Nielsen, J.R. Nyengaard, B. Pakkenberg, F.B. Sørensen, A. Vesterby, and M.J. West (1988) The new stereological tools: Disector, fractionator, nucleator and point sampled intercepts and their use in pathological research and diagnosis. APMIS *96*:857–881.

Holmes, A.H. (1927) Petrographic Methods and Calculations. London: Murby.

Howard, V., S. Ried, A. Baddeley, and A. Boyde (1985) Unbiased estimation of particle density in the tandem scanning reflected light microscope. J. Microsc. *138*:203–212.

Iniguez, C., M.J. Gayoso, and J. Carreres (1985) A versatile and simple method for staining nervous tissue using Giemsa dye. J. Neurosci. Methods *13*:77–86.

Iversen, L.L. (1987) Differences between early and late-onset Alzheimer's disease. Neurobiol. Aging *8*:554–555.

Krekule, I. and H.J.G. Gundersen (1989) Computer graphics enhancement of measures in stereology. Acta. Stereologica. *8*:533–535.

Kroustrup, J.P., and H.J.G. Gundersen (1983) Sampling problems in a heterogeneous organ: quantitation of relative and total volume of pancreatic islets by light microscopy. J. Microsc. *132*:43–55.

Lorente de Nó, R. (1934) Studies on the structure of the cerebral cortex. II. Continuation of the study of the ammonic system. J. Psychol. Neurol. (Lpz.) *46*:113–177.

Mani, R.B., J.B. Lohr, and D.V. Jeste (1986) Hippocampal pyramidal cells and aging in the human: A quantitative study of neuronal loss in sectors CA1 to CA4. Exp. Neurol. *94*:29–40.

Maragos, W.F., J.T. Greenamyre, J.B. Penney, and A.B. Young (1987) Glutamate dysfunction in Alzheimer's disease: An hypothesis. TINS *10*:65–67.

Matheron, G. (1971) The Theory of Regionalized Variables and Its Application. Les Cahiers du Centre de Morphologie Mathematique de Fontainebleau, No. 5. Ecole Nationale Superieure des Mines de Paris.

Pakkenberg, B., and H.J.G. Gundersen (1988) Total number of neurons and glial cells in human brain nuclei estimated by the disector and the fractionator. J. Microsc. *150*:1–20.

Pakkenberg, H., and J. Voigt (1964) Brain weight of Danes. Acta Anat. *56*:297–307.

Rosene, D.L., and G.W. Van Hoesen (1987) The hippocampal formation of the primate brain. In E.G. Jones and A. Peters (eds): Cerebral Cortex. Vol. 6, Further Aspects of Cortical Function, Including Hippocampus. New York: Plenum, pp. 345–456.

Saper, C.B. (1987) The value of alternative morphological approaches to Alzheimer's disease. Neurobiol. Aging *8*:576–577.

Scheff, S.W. (1987) Methodological considerations when assessing age related changes. Neurobiol. Aging *8*:571–573.

Sommer, W. (1880) Erkrankung des Ammonshorns als aetiologisches Moment der Epilepsie. Arch. Psychiatr. Nervenkr. *10*:631–675.

Stephan, H. (1975) Allocortex. In W. Bargmann (ed): Handbuch der Mikroskopischen Anatomie des Menschen, Bd 4: Nervensystem, Teil 9, Berlin: Springer, pp. 1–998.

Stephan, H. (1983) Evolutionary trends in limbic structures. Neurosci. Biobehav. Rev. *7*:367–374.

Sterio, D.C. (1984) The unbiased estimation of number and sizes of arbitrary particles using the disector. J. Microsc. *134*:127–136.

Swaab, D.F., and H.B.M. Uylings (1987) Density measures: Parameters to avoid. Neurobiol. Aging *8*:574–576.

West, M.J. (1990) A quantitative comparison of the hippocampal subdivisions of diverse species including hedgehogs, laboratory rodents, wild mice and men. Prog. Brain Res. *83*:13–36.

West, M.J., P.D. Coleman, and D.G. Flood (1988) Estimating the number of granule cells in the dentate gyrus with the disector. Brain Res. *448*:167–172.

Williams, R.W., and P. Rakic (1988) Three-dimensional counting: An accurate and direct method to estimate numbers of cells in sectioned material. J. Comp. Neurol. *278*:344–352.

Williams, R.W., and P. Rakic (1989) Erratum and Addendum. Three-dimensional counting: An accurate and direct method to estimate numbers of cells in sectioned material. J. Comp. Neurol. *281*:335.

Zola-Morgan, S., L.R. Squire, and D.G. Amaral (1986) Human amnesia and the medial temporal region: Enduring memory impairment following a bilateral lesion limited to field CA1 of the hippocampus. J. Neurosci. *6*:2950–2967.

APPENDIX A
Essential equipment

Point counting with linear arrays and neuron counting with disectors were carried out with an Olympus Video Stereological Analysis System manufactured by BICO, Copenhagen. This system consists of an Olympus BHS-2 microscope mounted with a Panasonic WV-CD 130 videocamera interfaced through a Commodore Amiga 2000 computer, equipped with a 2300 Genloc, to a Panasonic Color Video Monitor. Two programmable stepping motors control the x, y movement of the stage and a Heidenhain MT-2 microcator is mounted on the microscope to measure (to the nearest 0.5 μm) the up and down excursion of the stage. The point arrays or counting frames (created with the grid software provided with the system or commercially available programs like Delux Paint II) can be superimposed on the video images.

Although this and similar constellations of equipment (Curcio and Sloan, '86; Krekule and Gundersen, '89) expedite the analysis described here, they cannot be considered to be essential. Optical disectors and medium magnification point counting can be performed with simple projection microscopes (Gundersen et al., '88, Fig. 3). Only in the optical disector analysis of the tightly packed granule cell layer is the amount of light in projection systems a limiting factor. This problem can be dealt with to a certain degree by using thinner sections, larger sampling frames, and smaller disector heights than those used here, while maintaining the described sample volumes. The requirement for a microcator can be circumvented by using physical disectors based on photographs or drawings (West et al., '88) or, more efficiently, with double projections (Pakkenberg and Gundersen, '88). If optical disectors are to be used, it is essential that oil immersion optics be used so that the movement of the stage in the z-axis reflects the true movement of the optical plane through the section. The x, y movement of the stage can be carried out manually with the aid of the vernier scales present on most microscope stages.